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Review

Overview of Deadbeat Predictive Control Technology for Permanent Magnet Synchronous Motor System

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Abstract

Permanent magnet synchronous motors (PMSMs) have been widespread used in high-performance automation due to their superior control precision and energy efficiency. With the development of digital processors, deadbeat predictive control (DPC) has promising control strategy for PMSM drive systems. However, the performance of DPC is highly sensitive to inaccuracies in system parameters, while the system delay and total harmonic distortion can also significantly affect its control effectiveness. This paper presents a comprehensive review of the advantages of DPC strategies applied to PMSMs, as well as their limitations in practical applications. Several improvement approaches are also discussed, including parameter identification methods, robust control methods, and delay compensation methods. Finally, the paper explores future research trends in this field, highlighting potential directions for the development of predictive control in PMSM drive systems.

Keywords: deadbeat predictive control (DPC); permanent magnet synchronous motor (PMSM); predictive control; robust control

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1. Introduction

Electric motors currently constitute the largest portion of global electricity consumption. Improving motor efficiency has become crucial for energy conservation, environmental protection, and sustainable development worldwide. These factors are driving the global motor market toward high-efficiency solutions [1–9].

Permanent magnet synchronous motors (PMSMs) are characterized by their compact structure, high power factor, superior energy efficiency, and reliable performance, making them particularly suitable for various drive applications [10–17]. These motors have been extensively implemented in several critical fields, including but not limited to new energy vehicle propulsion systems, aerospace power systems, marine propulsion systems, and intelligent robotic systems [18–29]. The challenging operating conditions encountered in industrial environments necessitate the development of sophisticated control strategies to ensure optimal motor performance.

Conventional control methodologies for PMSM drive systems primarily include field-oriented control (FOC) [30–33] and direct torque control (DTC) [34–37]. While these approaches have demonstrated effectiveness, they exhibit certain limitations: FOC with PI regulators requires complex parameter tuning involving multiple intermediate variables and

shows performance degradation in high-dynamic scenarios, whereas DTC suffers from significant torque ripple at low speeds. These inherent constraints have motivated researchers to investigate advanced control strategies, including sliding mode control [38–42], position sensorless control [43–48], fault-tolerant control [49–55], and predictive control [56–62].

The discrete input signals of power electronic devices determine the switching state (ON/OFF) of power transistors, accelerating the development of discrete digital control technology. Meanwhile, due to the advancement of digital signal processing technology and the significant improvement in the computing power of processors, the research on predictive control methods has also received extensive attention.

Predictive control was initially proposed and implemented as early as the 1980s, with its specific characteristics varying significantly depending on the adopted prediction models and optimization algorithms. Predictive control strategies can be classified into two main categories: model predictive control (MPC) [32,63–68] and deadbeat predictive control (DPC) [69–72] based on their distinct prediction models and cost function. Among them, MPC can be divided into finite-control-set model predictive control (FCS-MPC) [32,63–67] and continuous-control-set model predictive control (CCS-MPC) [68].

Predictive control strategies have shown significant potential for improving the dynamic performance and operational stability of PMSMs, particularly in extreme operating environments. The accelerated evolution of contemporary control methodologies, power electronic devices, and computational hardware [6,73–79] has facilitated extensive research in this area and the development of novel predictive control methods. These emerging strategies exhibit enhanced prediction accuracy, improved reliability, and superior dynamic response characteristics. Table 1 lists the characteristics of each model.

Table 1. Summary table of feature comparisons among predictive algorithms.

Methods	Fixed Switching Rate	PWM Modulation Requirements	Algorithm Complexity	Whether Constraint Processing Is Required
Deadbeat Predictive Control	Yes	Yes	complicated	No
Continuous-set model predictive control	Yes	Yes	complicated	Yes
Finite-set model predictive control	No	No	normal	Yes

In predictive control, DPC has the advantages of fast response speed, high control accuracy, and small calculation time [80–84]. DPC is an advanced control strategy based on a discrete-time model. Its core concept involves directly computing the optimal voltage vector for the next control cycle by utilizing the system state and model information at the current moment. This ensures that the controlled variable accurately tracks the reference value within a single sampling period, thereby eliminating steady-state errors and achieving rapid dynamic responses. The fast dynamic response time and computational efficiency of DPC make it ideal for high-performance applications like electric vehicle drives and robot systems. Electric vehicle motors exhibit fast response characteristics under complex operating conditions. Particularly in terms of torque response time, the performance is significantly improved compared to traditional PI control, leading to enhanced transmission efficiency and energy utilization. In the field of robotics, the zero overshoot feature of DPC, when applied to speed controllers, enables precise joint control, effectively suppressing resonance and noise while achieving high control accuracy.

However, there are many problems in the application of DPC in PMSMs. For example, its prediction accuracy is extremely dependent on the accuracy of the model. When

parameter mismatch occurs, its prediction accuracy decreases, thereby leading to the decline in control performance [76,85–90]. To solve their existing problems, the current improvement measures and research directions are mainly shown in Figure 1.

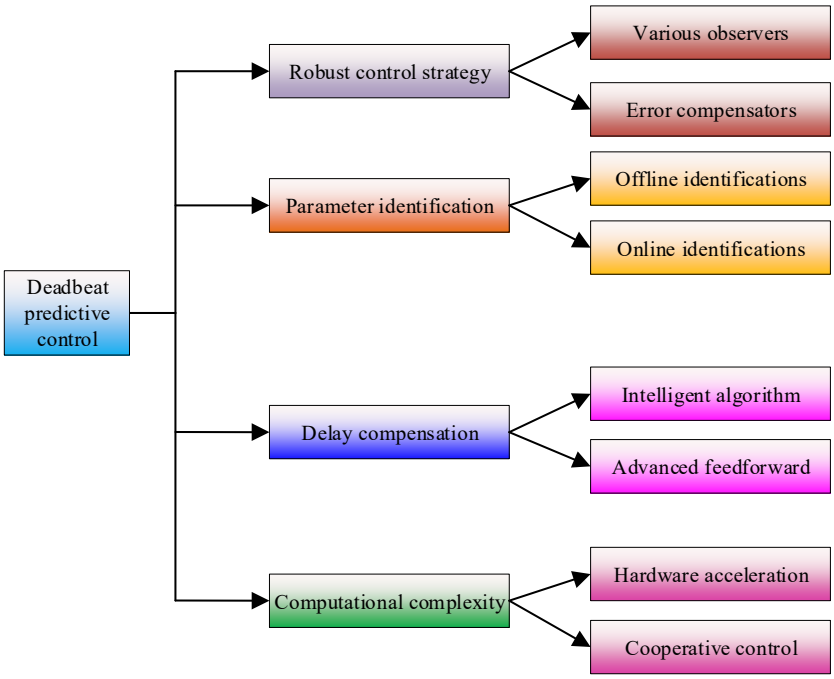


Figure 1. The main research direction of deadbeat predictive control.

The main work of this paper is to examine the challenges and enhancement strategies of the DPC method for permanent magnet synchronous motors from a technological development perspective. The key contribution lies in synthesizing the current research status of various improvement approaches. Fundamentally, it investigates and evaluates the feasibility of the DPC method while analyzing and summarizing the universality and efficacy of these techniques. Furthermore, the paper delves into the future development trends and potential research directions of the DPC method.

2. Composition of PMSM Drive System Based on Deadbeat Predictive Control

2.1. Composition of PMSM System

This section mainly introduces the model of PMSMs.

Figure 2 presents the circuit topology of the three-phase voltage source inverter used for permanent magnet synchronous motor drive. The inverter consists of six power switch devices, divided into two groups (upper arm and lower arm), with three switches in each group. During operation, the inverter uses pulse width modulation technology to sequentially control the switching states of the three-phase bridge arms. The complementary switching mode between the upper and lower arms generates eight possible combinations of switching vectors. Through this inverter process, the DC power supply is converted into a three-phase AC power supply to drive the PMSMs.

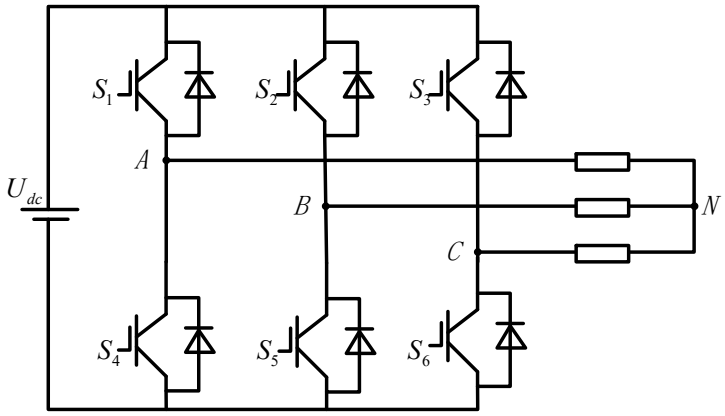


Figure 2. Three-phase PMSM voltage inverter topology.

The stator voltage equations of the PMSM can generally be expressed as

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \psi_f \end{cases} \tag{1}$$

where u_d and u_q are the d- and q-axis stator voltages, i_d and i_q are the d- and q-axis stator currents, respectively. L_d and L_q are the stator inductance of d- and q-axis, ψ_f is the winding resistance, ω_e is the machine's electrical rotor angular velocity, and R_s is the stator winding resistance.

The torque of PMSM can be expressed as

$$T_e = \frac{3}{2} P_n i_q \left[i_d (L_d - L_q) + \psi_f \right] \tag{2}$$

where P_n is the number of pole pairs of the motor.

The mechanical motion equation of the motor can be expressed as

$$J \frac{d\omega_m}{dt} = T_e - T_L - B\omega_m \tag{3}$$

where ω_m is the mechanical angular velocity of the motor, J is the moment of inertia, B is the friction coefficient, and T_L is the load torque.

2.2. Conventional DPCC Model

For the deadbeat predictive current control (DPCC), this approach originates from discrete linear state feedback control, enabling the feedback current to track the reference current within a finite time. Its principle involves obtaining the motor current through sampling, performing coordinate transformation, and comparing it with the reference current. The resulting difference is then processed by a non-differential current controller. The output voltage undergoes inverse coordinate transformation and subsequently generates an inverter switch pulse signal via the Space Vector Pulse Width Modulation (SVPWM) module, as illustrated in Figure 3 below. Compared with conventional current controllers, the proportional–integral (PI) controller has been replaced by a deadbeat controller, which features constant switching frequency, rapid dynamic response, high bandwidth, minimal current ripple, and ease of implementation. Under a predictive current control framework, the motor dynamics are discretized via the forward Euler approximation [91], the discretized voltage equation can be rewritten as

$$\begin{cases} u_d(k) = R_s i_d(k) + (L_d / T_s)[i_d(k+1) - i_d(k)] - \omega_e L_q i_q(k) \\ u_q(k) = R_s i_q(k) + (L_q / T_s)\left[i_q(k+1) - i_q(k)\right] + \omega_e L_d i_d(k) + \omega_e \psi_f \end{cases} \tag{4}$$

where T_s is the sampling time.

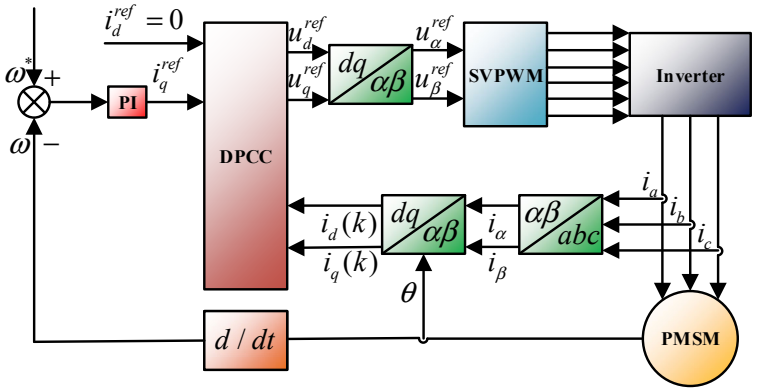


Figure 3. The control diagram of conventional DPCC.

By replacing the d- and q-axis currents at $(k+1)$-th in (4) with the reference current, the voltage model of the conventional discrete DPCC method can be expressed as

$$\begin{cases} u_d^{\text{pre}}(k) = R_s i_d(k) + (L_d/T_s)\left[i_d^{\text{ref}}(k) - i_d(k)\right] - \omega_e L_q i_q(k) \\ u_q^{\text{pre}}(k) = R_s i_q(k) + (L_q/T_s)\left[i_q^{\text{ref}}(k) - i_q(k)\right] + \omega_e L_d i_d(k) + \omega_e \psi_f \end{cases} \tag{5}$$

2.3. Conventional DPSC Model

DPCC can enable the current loop to achieve the fastest dynamic performance and reach the reference value without overshoot. However, DPCC is only used in the current inner loop of variable-speed drive systems, and the responses of speed and torque will still be limited by the PI controller, which is typically used in the outer speed loop of the above-mentioned method.

The PI controller can realize perfect tracking capability between the speed reference value and the speed feedback value. The output of PI controller can be expressed as

$$i_q^{ref} = K_p(\omega_m^{ref} - \omega_m^*) + K_i \int K_p(\omega_m^{ref} - \omega_m^*)dt \tag{6}$$

where K_p is the proportional coefficient of the PI controller, and K_i is the integral coefficient of the PI controller.

In order to further improve the dynamic characteristics of the system, deadbeat predictive control is also introduced into the design of the speed loop (DPSC), as shown in Figure 4. The speed loop Equation (3) by using Euler method can be discretized as

$$\omega_m^p(k) = A\omega_m(k-1) + Bi_q(k-1) + C(k) \tag{7}$$

where T_{sp} is the sampling time of speed loop. $A = 1 - \frac{T_{sp}B}{J}$, $B = \frac{3P_n\psi_f T_{sp}}{2J}$, $C(k) = -\frac{T_L T_{sp}}{J}$.

According to (7), the output reference current value of the speed loop can be expressed as

$$i_q(k) = \frac{2J}{3P_n\psi_f T_{sp}}\left[\omega_m^{ref} - (1 - \frac{T_{sp}B}{J})\omega_m(k) + \frac{T_L T_{sp}}{J}\right] \tag{8}$$

Both DPCC and DPSC can achieve high dynamic response and reach the reference value without overshoot. However, the control effects of the two will decline sharply especially when the parameters are mismatched. Based on the brief introduction and analysis of the above theories, it is evident that they exhibit several limitations. With the widespread application of DPC, an increasing number of researchers have focused

on improving the performance of motor drive systems. The next chapter will provide a detailed discussion of existing research methodologies.

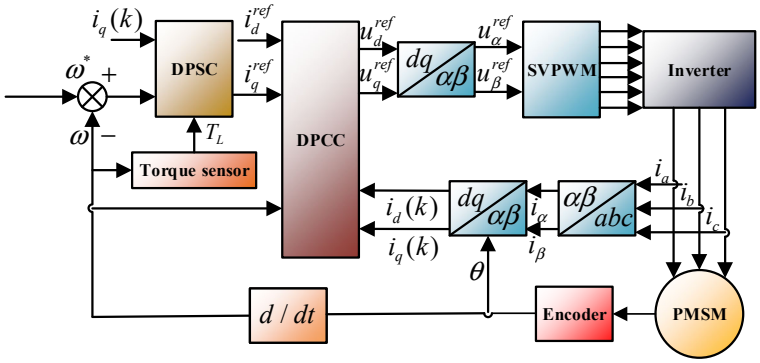


Figure 4. The combination of DPSC and DPCC.

3. An Overview of the Recent Development for Deadbeat Predictive Control Methods of PMSM

To better understand the development of different methods, identify more advantages of DPC, and explore its future directions, we introduce and analyze relevant approaches. These methods are selected from experimentally validated studies to allow for a quantitative assessment of their performance.

3.1. Deadbeat Predictive Current Control of PMSM

In discrete-time PMSM drive systems, sampling delays, inverter delays, and control system delays collectively degrade the prediction accuracy of DPCC. Furthermore, due to the algorithm's heavy reliance on mathematical models, its robustness has become a subject of extensive research [91–98].

3.1.1. Robust Control Strategy

The accuracy of motor parameters (such as inductance, stator resistance, and flux linkage) directly affects the control performance of motor drive system. However, motor parameters will change due to factors such as temperature rise, magnetic saturation, and aging, resulting in model mismatch, which in turn affects the current tracking accuracy and system stability. Therefore, how to improve the robustness of the DPCC scheme to ensure the efficient dynamic and steady-state performance of DPCC has become a current research hotspot [36,99–108].

A disturbance rejection strategy can be effectively adopted to substantially improve the robustness of the DPCC method. The widely studied approach involves the Extended State Observer (ESO) [109–114] and Sliding Mode Observer (SMO) [115–122]. Specifically, as shown in Figure 5, the ESO treats model uncertainties, load disturbances, and other unmodeled dynamics as a lumped disturbance, which is estimated in real time and actively compensated, thereby enhancing the system's ability to reject external disturbances. Next, take surface-mounted PMSM as an example. The state space equation of the d- and q-axis loop can be expressed as

$$\begin{bmatrix} \frac{di_d}{dt} \\ \frac{di_q}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L_s} & \omega_e \\ -\omega_e & -\frac{R_s}{L_s} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \end{bmatrix} \begin{bmatrix} u_d \\ u_q \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{\omega_e \psi_f}{L_s} \end{bmatrix} \tag{9}$$

Rewrite (9) as the standard form of the state space equation:

$$\dot{x}_{dq}(t) = A_{dq}(t)x_{dq}(t) + B_{dq}u_{dq}(t) + C_{dq}(t) \tag{10}$$

where $\dot{x}_{dq}(t) = \begin{bmatrix} \frac{di_d}{dt} \\ \frac{di_q}{dt} \end{bmatrix}$, $x_{dq}(t) = \begin{bmatrix} i_d \\ i_q \end{bmatrix}$, $u_{dq}(t) = \begin{bmatrix} u_d \\ u_q \end{bmatrix}$, $A_{dq}(t) = \begin{bmatrix} -\frac{R_s}{L_s} & \omega_e \\ -\omega_e & -\frac{R_s}{L_s} \end{bmatrix}$, $B_{dq} = \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \end{bmatrix}$, $C_{dq}(t) = \begin{bmatrix} 0 \\ -\frac{\omega_e \psi_f}{L_s} \end{bmatrix}$.

When considering the effects of parameter mismatch, (9) can be rewritten as

$$\begin{bmatrix} \frac{di_d}{dt} \\ \frac{di_q}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R_s+\Delta R_s}{L_s+\Delta L} & \omega_e \\ -\omega_e & -\frac{R_s+\Delta R_s}{L_s+\Delta L} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \frac{1}{L_s+\Delta L} & 0 \\ 0 & \frac{1}{L_s+\Delta L} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{\omega_e(\psi_{f1}+\Delta\psi_f)}{L_s+\Delta L} \end{bmatrix} \tag{11}$$

where ΔL and $\Delta\Psi_f$ are the difference between the actual value and the nominal value of the motor parameters.

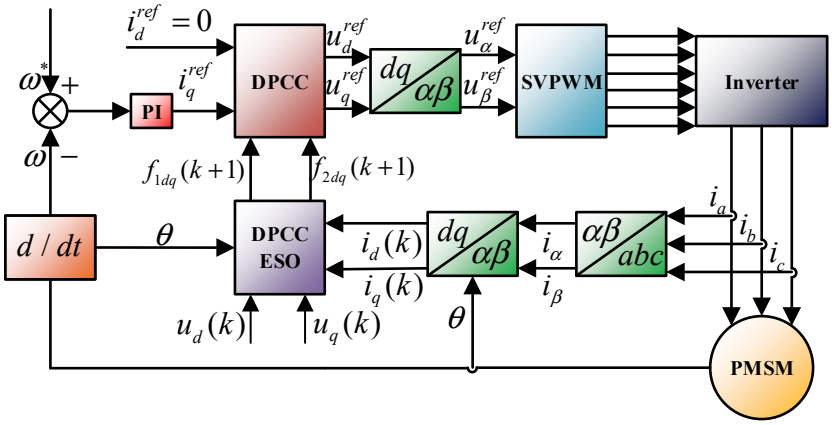


Figure 5. The control diagram of DPCC-ESO.

It can be observed that when multiple parameter mismatches occur in the motor, the motor equations become extremely complex, making it highly challenging to simultaneously monitor the variations in all parameters. Based on the concept of ESO, the disturbances caused by parameter mismatches can be equivalently treated as an additional input. Thus, the state space representation of (11) can be expressed as

$$\dot{x}_{dq}(t) = A_{dq}(t)x_{dq}(t) + B_{dq}u_{dq}(t) + C_{dq}(t) + B_{dq}u_{dq}^{dis}(t) \tag{12}$$

where $u_{dq}^{dis}(t) = \begin{bmatrix} u_d^{dis}(t) \\ u_q^{dis}(t) \end{bmatrix}$, $u_d^{dis}(t)$ is the disturbance caused by parameter mismatch in the d-axis, and $u_q^{dis}(t)$ is the disturbance caused by parameter mismatch in the q-axis.

Therefore, the disturbance caused by the parameter mismatch can be equivalent to the voltage disturbance, and the ESO of the d- and q-axis can be expressed as

$$\begin{cases} \dot{f}_{1dq}(t) = A_{dq}(t)x_{dq}(t) + B_{dq}u_{dq}(t) + C_{dq}(t) + B_{dq}f_{2dq}(t) + \alpha_1[x_{dq}(t) - f_{1dq}(t)] \\ \dot{f}_{2dq}(t) = \alpha_2[x_{dq}(t) - f_{1dq}(t)] \end{cases} \tag{13}$$

where $f_{1dq}(t)$ is the d- and q-axis current estimated by the observer, $f_{2dq}(t)$ is the voltage disturbance $u_{dq}^{dis}(t)$ of the d- and q-axis estimated by the observer, and α_1 and α_2 are the gain of the observer.

It can be seen from (13) that when the gain of the selected observer is greater than 0, the observer can remain convergent for a long time. For example, if the d- and q-axis current $f_{1dq}(t)$ observed by ESO is less than the actual d- and q-axis current $x_{dq}(t)$, it can be seen from (13) that $\alpha_2\left[x_{dq}(t) - f_{1dq}(t)\right]$ is greater than 0, and then the d- and q-axis voltage disturbance $f_{2dq}(t)$ observed by ESO will also increase, thus increasing $\dot{f}_{1dq}(t)$. At the same

time, the $\alpha_1\left[x_{dq}(t)-f_{1dq}(t)\right]$ term of $\dot{f}_{1dq}(t)$ will also increase $\dot{f}_{1dq}(t)$, and under the above conditions, $\dot{f}_{1dq}(t)$ will also increase, thereby increasing $f_{1dq}(t)$ until $x_{dq}(t)=f_{1dq}(t)$.

When the gain of the observer is selected appropriately, $f_{1dq}(t)$ will converge to $x_{dq}(t)$, that is the d- and q-axis current estimated by the observer will be equal to the actual d- and q-axis current value; $f_{2dq}(t)$ will converge to $u_{dq}^{dis}(t)$, that is the d- and q-axis voltage disturbance estimated by the observer will be equal to the actual d- and q-axis voltage disturbance.

In the actual motor control system, DSP is generally used, so the ESO needs to be discretized. After the discretization of (13), the d- and q-axis ESO can be expressed as

$$\begin{cases} f_{1dq}(k+1)=T_sA_{dq}(k)x_{dq}(k)+T_sB_{dq}u_{dq}(k)+T_sC_{dq}(k)+T_sB_{dq}f_{2dq}(k) \\ +T_s\alpha_1x_{dq}(k)+(1-T_s\alpha_1)f_{1dq}(k) \\ f_{2dq}(k+1)=T_s\alpha_2[x_{dq}(k)-f_{1dq}(k)]+f_{2dq}(k) \end{cases} \tag{14}$$

where $A_{dq}(k)$, $C_{dq}(k)$, $x_{dq}(k)$, and $u_{dq}(k)$ are the sample values of $A_{dq}(t)$, $C_{dq}(t)$, $x_{dq}(t)$, and $u_{dq}(t)$ at k time, respectively, $f_{1dq}(k)$ and $f_{1dq}(k+1)$ are the estimates of the d- and q-axis current output by the d- and q-axis observer at k time and $(k+1)$ time, respectively.

The predictive current can be expressed as

$$\begin{bmatrix} i_d(k+2) \\ i_q(k+2) \end{bmatrix}=T_sA_{dq}(k)\begin{bmatrix} f_{1d}(k+1) \\ f_{1q}(k+1) \end{bmatrix}+T_sC_{dq}(k)+T_sB_{dq}\begin{bmatrix} u_d(k+1)+f_{2d}(k+1) \\ u_q(k+1)+f_{2q}(k+1) \end{bmatrix} \tag{15}$$

where $A_{dq}(k)$ and $C_{dq}(k)$ are supposed to be $A_{dq}(k+1)$ and $C_{dq}(k+1)$, respectively. However, the time-varying quantities in the matrices $A_{dq}(k+1)$ and $C_{dq}(k+1)$ are $\omega_e(k+1)$ and $\theta_e(k+1)$, respectively, due to the large mechanical time constant of the motor, it can be considered that $\omega_e(k+1)$ is about equal to $\omega_e(k)$ and $\theta_e(k+1)$ is about equal to $\theta_e(k)$. So $A_{dq}(k+1)\approx A_{dq}(k)$ and $C_{dq}(k+1)\approx C_{dq}(k)$.

According to the principle of DPCC, the current is expected to reach the given value at $(k+2)$ time, so the voltage reference value at $(k+1)$ time can be calculated, and the voltage reference value at $(k+1)$ time can be expressed as

$$\begin{bmatrix} u_d^*(k+1) \\ u_q^*(k+1) \end{bmatrix}=\frac{B_{dq}^{-1}}{T_s}\begin{bmatrix} i_d^* \\ i_q^* \end{bmatrix}-B_{dq}^{-1}A_{dq}(k)\begin{bmatrix} f_{1d}(k+1) \\ f_{1q}(k+1) \end{bmatrix}-B_{dq}^{-1}C_{dq}(k)-\begin{bmatrix} f_{2d}(k+1) \\ f_{2q}(k+1) \end{bmatrix} \tag{16}$$

Figures 6 and 7 show the performance of DPCC-ESO under parameter mismatch. It can be seen that DPCC-ESO can effectively suppress the current pulsation caused by inductance mismatch. Even if flux linkage mismatch occurs, ESO can still enable the system to accurately observe the actual current.

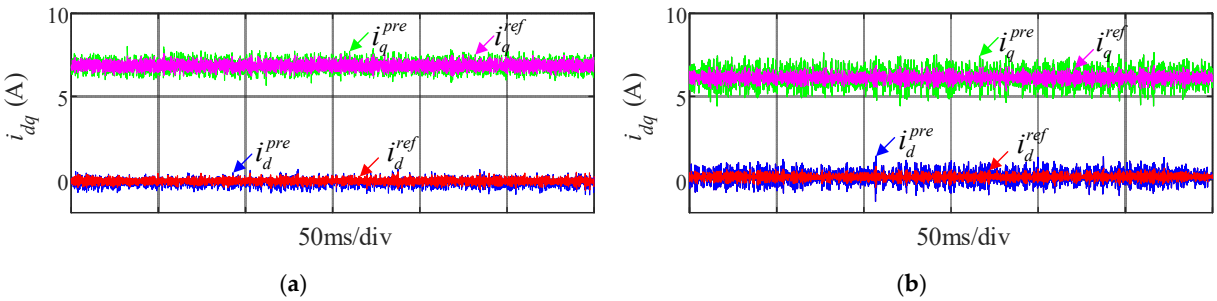


Figure 6. Current waveforms with a speed of 1000 rpm under rated load of 5 Nm when inductance mismatch occurs. (**a**) DPCC-ESO, (**b**) DPCC.

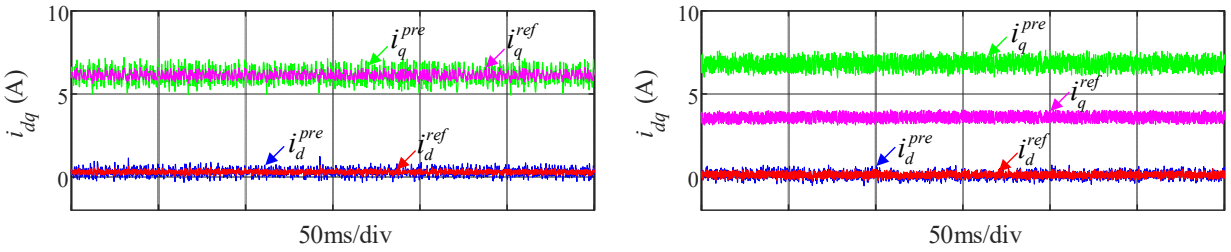


Figure 7. Current waveforms with a speed of 1000 rpm under rated load of 5 Nm when Flux linkage mismatch occurs. (**a**) DPCC-ESO, (**b**) DPCC.

The second method to enhance the robustness of DPCC is based on various observers. To further mitigate the impact of parameter mismatch, recent studies [123–128] have extensively investigated the ultra-local model (ULM)-based DPCC method. In this approach, the current controller is designed based on the ultra-local model rather than the conventional PMSM model. The ULM requires only two tunable gains: the controller proportional gain and the disturbance observation gain. Notably, when the proportional gain of the ULM-based controller is set to the reciprocal of the inductance, the system exhibits superior dynamic performance. Moreover, the ULM treats the back-EMF and resistive voltage drop of the PMSM as lumped disturbances, which are estimated via a disturbance observer. This formulation inherently reduces the controller's dependence on the accurate knowledge of the resistance and back-EMF parameters.

In [123], the disturbance term was estimated using an extended state observer (ESO), while the controller proportional gain was directly set as the reciprocal of the inductance. This indicates that the method still relies on the PMSM's inductance parameter. To further improve this approach, ref. [127] proposed an effective parameter-free ultra-local model-based DPCC method where the reciprocal of inductance was estimated using a finite-time gradient method and the disturbance term was observed via ESO. Paper [128] introduced a model-free DPCC with ESO based on parameter estimation, capable of simultaneously estimating both model gain and disturbance terms. While the methods in [127,128] eliminate the need for motor parameters, the observation speed of disturbance terms remains constrained by the observer bandwidth limitations. During dynamic processes, since the disturbance term continuously varies and the observer cannot instantly achieve unbiased estimation, observer-based robust DPCC approaches face challenges in achieving the high dynamic performance characteristic of ideal DPCC. Therefore, in order to achieve the high dynamic performance of the ideal DPCC, motor parameters are indispensable.

Unlike the method of avoiding the use of motor parameters, another solution to enhance robustness is to obtain accurate machine parameters through online parameter estimation, The widely studied conventional PMSM parameter identification methods mainly include recursive least square method (RLS), Extended Kalman Filter (EKF), and signal injection, etc. [129–134]. In [129], a motor parameter identification strategy by genetic factor least square method was used to adjust the controller parameter of DPCC method. In [131], an online parameter identification approach utilizing the Kalman filter algorithm for real-time parameter estimation was developed. In [132], a Lyapunov function was utilized to suppress current tracking errors, enforcing its time derivative to be negative-definite for stability. Although these conventional methods have the characteristics of being easy to implement and having low measurement noise, they also have problems such as low accuracy, the need for additional conditions, and high computational complexity. Table 2 lists the advantages and disadvantages of these several conventional methods.

Table 2. Comparison of conventional PMSM parameter identification methods.

Methods	Advantage	Disadvantage	Complexity	Accuracy
RLS	Easy to implement	Low accuracy	High	Medium
EKF	Strong robustness to noise	Complex computations	High	High
MRAS	Wide speed range and easy to implement	Need additional conditions	Medium	Slightly High

Parameter identification based on signal injection [102,135–138] is also a widely used method. Paper [135] proposed an online parameter identification method based on the BKF, which decouples the inductance and permanent magnet flux linkage for independent parameter estimation. This approach reduces computational complexity while maintaining high identification accuracy, significantly improving performance under varying load conditions. Two effective signal injection methods are shown in Figure 8. Paper [136] proposed a full parameter identification method for PMSM based on high-frequency signal injection. By combining high-frequency signal injection with the DPCC method, it effectively solved the problem that the conventional parameter identification method could not perform full parameter identification due to the lack of rank in the motor equations. Paper [137] proposed a full parameter identification method for PMSM under steady-state conditions based on triangular wave signal injection. Similar to the method based on high-frequency sinusoidal signal injection, due to the injection of new signals, it effectively solves the problem of missing order in the conventional motor equation, and at the same time, the bandwidth requirement for the controller is not very high. Although the identification effect of such identification methods is very good, the identification time is also very long. Therefore, the parameter identification method based on signal injections is not applicable to servo motor systems with frequent changes in rotational speed and load.

Another method for using motor parameters is the parameter identification method based on error correction [94,139,140]. In the ESO-based parameter-free ultra-local model DPCC method, when parameter mismatch occurs in the system, the disturbance value observed by the ESO becomes non-zero, and vice versa. Building upon this principle, recent studies [94,140] have proposed error-correction-based parameter identification methods. These approaches continuously adjust and correct the controller's estimated motor parameters based on the ESO-observed disturbance values until they converge to zero. The disturbance observation value of ESO being zero indicates perfect alignment between the controller's estimated parameters and the actual motor parameters, eliminating any parameter deviation in the control system. This enables the realization of ideal DPCC with high dynamic performance. However, due to the relatively complex design of the observer, the parameter identification is also relatively complicated.

To address the shortcomings of the observer, a control method that compensates for current errors can effectively solve this problem [141–144], The error-correction-based parameter identification methods proposed in [141,143] can perform parameter identification during both dynamic and steady-state processes, making them particularly suitable for servo applications with frequent operating condition changes. In [143], a predictive current error compensation-based strong robust method for PMSM drive systems is proposed. By constructing two new variables (N and M), M involves the inductance parameter, and N involves the flux linkage parameter. Adjust the parameters N and M based on the error of the predictive current. When there is no error between the predicted current and the actual current, it indicates that N and M have reached accurate parameters. This approach is computationally efficient and eliminates the need for complex coefficient adjustments. Furthermore, it reduces system memory requirements by avoiding the implementation of conventional observers or intricate parameter identification algorithms. In [141], a model

parameter self-correcting deadbeat predictive current control method (MPSC-DPCC) is proposed. This article proposes a model parameter self-calibration scheme, which uses motor data under different operating conditions to form a new motor equation. The main idea is to determine whether the current difference between two cycles is large enough. If the current difference is large enough, it is equivalent to the motor being in two different operating conditions. Furthermore, the prediction error data, current data, and rotational speed data under these two working conditions were recorded. Then, the least square method was used for identification and combined with ESO to establish a closed-loop disturbance regulation system. By adjusting the model parameters, the observed disturbance was adjusted to zero, and the identification of the three parameters of motor inductance, resistance, and flux linkage could be achieved without signal injection. While addressing the disadvantage of [143], that it cannot perform full parameter identification, it ensures the robustness of DPCC, thereby achieving rapid dynamic performance without overshoot. The implementation complexity of the proposed methods is listed in Table 3.

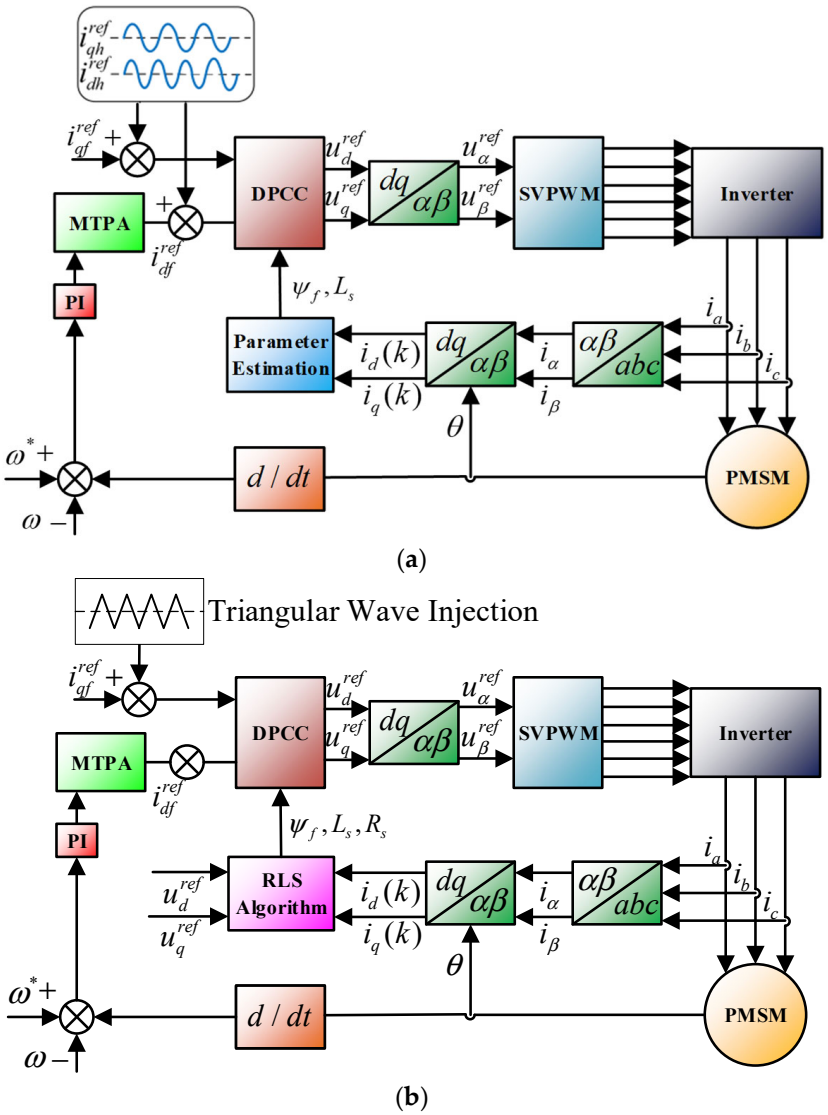


Figure 8. (**a**) high-frequency signal injection; (**b**) triangular wave signal injection.

Table 3. Implementation complexity of the proposed methods.

Method	DPCC	DPCC-ESO	Method in [Ultra]	Method in [CEC]	Method in [Wang]
Clock period	5466	5682	5548	6224	6118
Algorithm execution time	36.54 μs	37.98 μs	36.84 μs	42.68 μs	42.04 μs

3.1.2. Delay Compensation

Under ideal conditions, as shown in Figure 9a, the motor controller would instantaneously apply the computed voltage vector to the motor following the acquisition of motor data. However, in practical applications, the process of AD conversion and subsequent voltage vector calculation consumes a certain amount of time in Figure 9b. Consequently, the calculated voltage vector cannot be applied to the motor immediately but rather at the next control cycle, resulting in a one-step delay within the control system. Without proper delay compensation, this latency can adversely affect the overall control performance of the system [145–148].

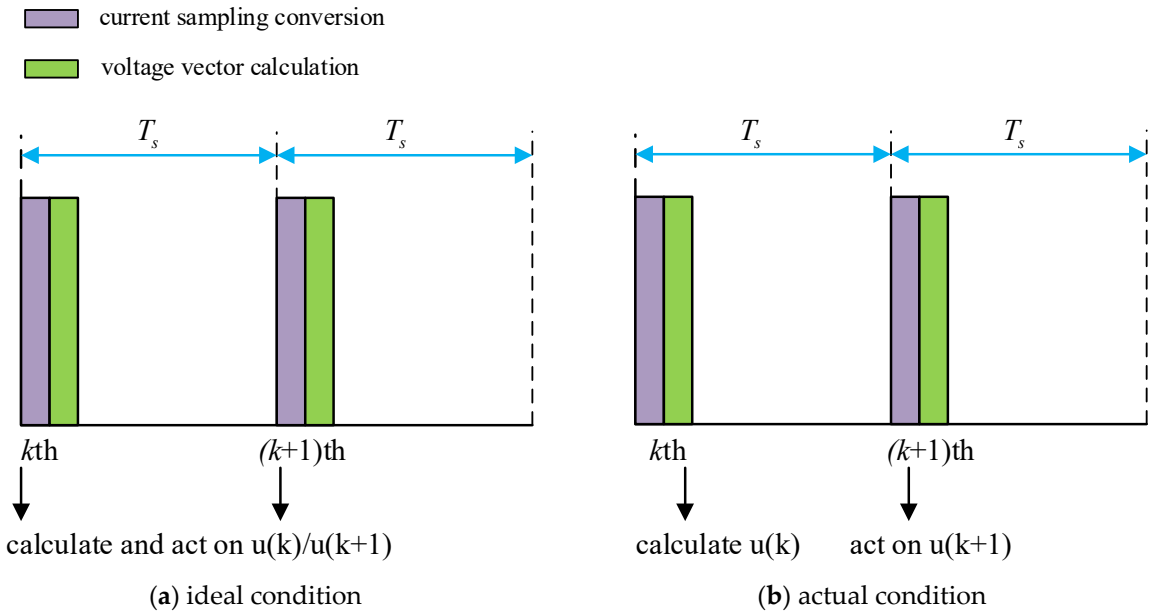


Figure 9. The timing diagrams of DPC under ideal and actual conditions.

As can be concluded from the above analysis, the voltage vector calculated during the k-th control cycle can only be applied to the motor at the beginning of the (k + 1)-th control cycle. To perform delay compensation, the current value at the (k + 1)-th control cycle should be calculated based on the current at k-th control cycle and the voltage vector acting on the motor within the current period at the k-th control cycle. Paper [149] proposes a voltage vector compensation strategy to mitigate the computational delay in the inverter system. This approach effectively reduces current distortion caused by system delays while enhancing control precision. The control timing diagram is shown in Figure 10 below.

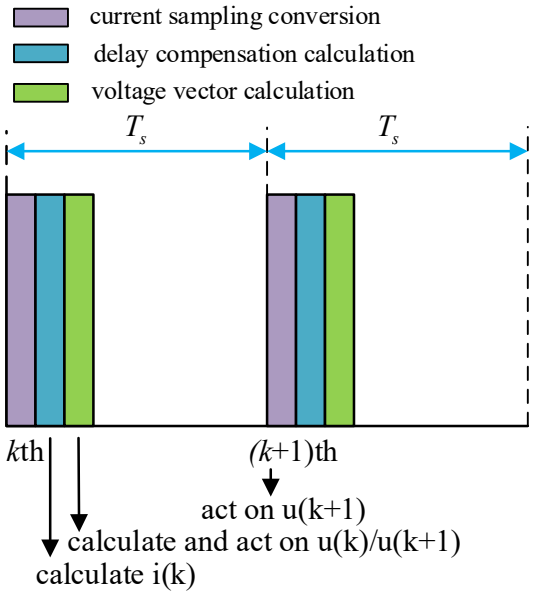


Figure 10. Control timing diagram with delay compensation.

The control diagram of DPCC with one-step delay is shown in Figure 11 and the one-step delay compensation can be expressed as

$$\begin{bmatrix} i_d^{\text{com}}(k+1) \\ i_q^{\text{com}}(k+1) \end{bmatrix} = F \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix} + G \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix} + H \tag{17}$$

where $F = \begin{bmatrix} 1 - R_s \frac{T_s}{L_s} & T_s \omega_e \frac{L_d}{L_s} \\ -T_s \omega_e \frac{L_d}{L_s} & 1 - R_s \frac{T_s}{L_s} \end{bmatrix}$, $G = \begin{bmatrix} \frac{T_s}{L_s} & 0 \\ 0 & \frac{T_s}{L_s} \end{bmatrix}$. $H = \begin{bmatrix} 0 \\ -\omega_e \psi_f \frac{T_s}{L_s} \end{bmatrix}$.

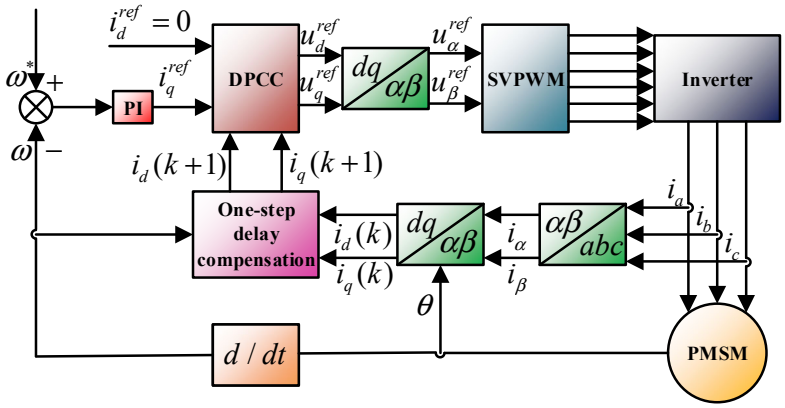


Figure 11. The control diagram of DPCC with delay compensation.

By substituting (17) into (5), the DPCC method considering one-step delay compensation can be expressed as

$$\begin{cases} u_d^{pre}(k) = R_s i_d^{\text{com}}(k+1) + (L_s/T_s)\left[i_d^{ref}(k) - i_d^{\text{com}}(k+1)\right] \\ -\omega_e L_s i_q^{com}(k+1) \\ u_q^{pre}(k) = R_s i_q^{\text{com}}(k+1) + (L_s/T_s)\left[i_q^{ref}(k) - i_q^{\text{com}}(k+1)\right] \\ +\omega_e L_s i_d^{\text{com}}(k+1) + \omega_e \psi_f \end{cases} \tag{18}$$

In [149,150], considering the inverter output delay and current sampling delay, a deadbeat multi-step predictive compensation scheme is proposed. Based on the measured

current and voltage values at time k, a multi-step prediction is performed to obtain the reference voltage at (k + 2)-th, as expressed in Equation (18). After the first step of prediction is completed, the predicted current value at time (k + 1)-th is stored in the register. Subsequently, the second-step prediction is carried out using the reference current, motor inductance, resistance, flux linkage, and other relevant parameters to calculate the voltage value u(k + 2) at (k + 2)-th. This approach effectively compensates for the delay inherent in the current loop and enhances the bandwidth of the current control system [151].

3.1.3. DPCC Based on Incremental Model

The parameter mismatch of PMSM can seriously affect the control performance of the motor [76,87,152]. Specifically, inductance mismatch can lead to high-frequency noise and harmonics, flux linkage mismatch can cause steady errors, while resistance mismatch has a relatively small impact on the control performance of the motor [153]. Incremental model can eliminate the problem of steady-state error caused by flux linkage mismatch in DPCC method [154]. The control principle of the incremental DPCC method is shown in Figure 12.

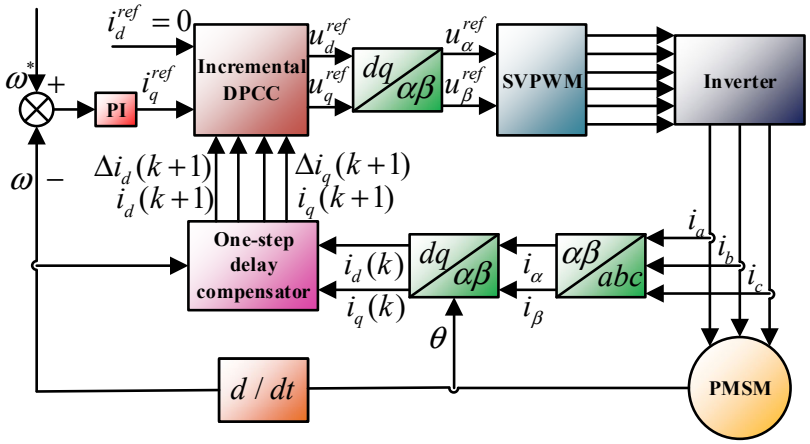


Figure 12. IDPCC with one-step delay compensation.

The predicted current of incremental DPCC method at (k + 1)-th can be expressed as

$$\begin{bmatrix} i_d^{pre}(k+1)-i_d(k) \\ i_q^{pre}(k+1)-i_q(k) \end{bmatrix} = \begin{bmatrix} 1-\frac{T_sR_s}{L_s} & T_s\omega_e(k) \\ -T_s\omega_e(k) & 1-\frac{T_sR_s}{L_s} \end{bmatrix} \begin{bmatrix} i_d(k)-i_d(k-1) \\ i_q(k)-i_q(k-1) \end{bmatrix} + \frac{T_s}{L_s} \begin{bmatrix} u_d(k)-u_d(k-1) \\ u_q(k)-u_q(k-1)+\frac{T_s\psi_f}{L_s}[-\omega_e(k)+\omega_e(k-1)] \end{bmatrix} \quad (19)$$

In incremental predictive model, the back electromotive force terms (back-EMF) at k-th and ($k-1$)-th can be considered identical, (11) can be further rewritten as

$$\begin{bmatrix} i_d^{pre}(k+1)-i_d(k) \\ i_q^{pre}(k+1)-i_q(k) \end{bmatrix} = \begin{bmatrix} 1-\frac{T_sR_s}{L_s} & T_s\omega_e(k) \\ -T_s\omega_e(k) & 1-\frac{T_sR_s}{L_s} \end{bmatrix} \begin{bmatrix} i_d(k)-i_d(k-1) \\ i_q(k)-i_q(k-1) \end{bmatrix} + \frac{T_s}{L_s} \begin{bmatrix} u_d(k)-u_d(k-1) \\ u_q(k)-u_q(k-1) \end{bmatrix} \quad (20)$$

According to (20), it can be seen that the incremental DPCC has already demonstrated the influence of the flux linkage parameter, the calculation formula of the expected reference voltage is derived as

$$\begin{bmatrix} \Delta u_d^{ref}(k+1) \\ \Delta u_q^{ref}(k+1) \end{bmatrix} = \frac{L_s}{T_s} \begin{bmatrix} \Delta i_d^{ref}(k)-\Delta i_d^{pre}(k+1) \\ \Delta i_q^{ref}(k)-\Delta i_q^{pre}(k+1) \end{bmatrix} + \begin{bmatrix} R_s & -L_s\omega_e(k) \\ L_s\omega_e(k) & R_s \end{bmatrix} \begin{bmatrix} \Delta i_d^{pre}(k+1) \\ \Delta i_q^{pre}(k+1) \end{bmatrix} \quad (21)$$

The reference voltages of the d- and q-axis can be expressed as

$$\begin{bmatrix} u_d^{ref}(k+1) \\ u_q^{ref}(k+1) \end{bmatrix} = \begin{bmatrix} u_d^{ref}(k) \\ u_q^{ref}(k) \end{bmatrix} + \frac{L_s}{T_s} \begin{bmatrix} \Delta i_d^{ref}(k) - \Delta i_d^{pre}(k+1) \\ \Delta i_q^{ref}(k) - \Delta i_q^{pre}(k+1) \end{bmatrix} + \begin{bmatrix} R_s & -L_s\omega_e(k) \\ L_s\omega_e(k) & R_s \end{bmatrix} \begin{bmatrix} \Delta i_d^{pre}(k+1) \\ \Delta i_q^{pre}(k+1) \end{bmatrix} \quad (22)$$

Figure 13 shows the performance of DPCC and IDPCC under flux linkage mismatch. From the above equations and Figure 13, it can be seen that the incremental model does not contain flux linkage terms. Therefore, this method can effectively eliminate the influence of flux linkage errors on the current. Although the influence of flux linkage error has been eliminated, the current static error caused by the mismatch of resistance and inductance parameters still exists.

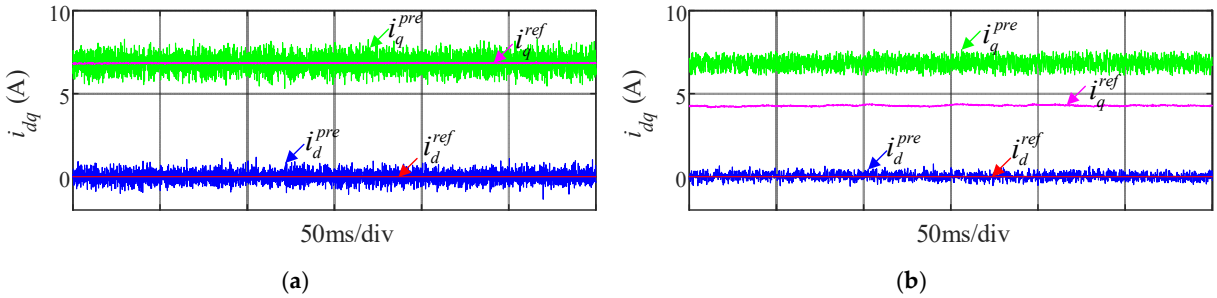


Figure 13. Current waveforms with a speed of 1000 rpm under rated load of 5 Nm when flux linkage mismatch occurs. (**a**) IDPCC, (**b**) DPCC.

To address the system disturbances and current pulsations caused by inaccurate inductance parameter in IDPCC, ref. [155] combines the incremental predictive model with the Romberg observer to compensate for motor disturbances, further eliminating current error and effectively reducing current fluctuations. Paper [154] has proposed a robustness optimization algorithm based on IDPCC (RIDPCC), which mainly enhances the robustness against inductance mismatch and load disturbances. It can operate normally without rotor flux linkage. Through multi-step prediction of RIDPCC, the algorithm's anti-interference ability against load disturbances has been improved. Finally, the incremental model was combined with ESO to estimate the error caused by the inductance mismatch of the estimator and compensated for it in the predictive model. In [156], a predictive current control method based on a self-correcting incremental model is introduced. To address the issue where stator inductance still needs to be accounted for in the prediction model, a self-tuning incremental model-based predictive current control (STIM-PCC) method is proposed. This method leverages the prediction error caused by inductance mismatch to adjust the nominal inductance value and adaptively update the incremental model, thereby achieving accurate predictive control performance. Additionally, a voltage-constrained compensation approach is presented, which effectively mitigates current drop during dynamic processes. Compared with conventional compensation techniques, the proposed method significantly improves parameter robustness and enables timely detection and correction of inductance mismatches. In view of the different characteristics of the *d*- and *q*-axis prediction model, paper [157] proposed a simple predictive current control without motor parameters. By using an incremental model in the *q*-axis predictive equation to extract the resistance and flux linkage of the model to investigate the influence of parameters on control performance, an estimation factor was designed in the *d*-axis prediction equation to replace the inductance term in the predictive model and was calculated based on a simple rhythm in real time. A prediction model without motor parameters was finally established. However, the long-standing problem with IDPCC is that it will increase the steady-state fluctuation in current and total harmonic distortion (THD). How to reduce the THD of IDPCC is also a major research direction in the future.

3.1.4. High Performance of DPSC

DPSC has been developed for variable-speed drive systems [158,159], where the current loop and speed loop are modeled as an integrated structure rather than a cascaded one. In contrast to the cascaded architecture, the control voltage is directly computed by the predictive system model. Although the internal current loop is excluded, the cost function of DPSC remains highly complex due to the inclusion of numerous weighting factors [159]. This increases the dimensionality of the prediction model's state equation, raises computational demands, and complicates parameter tuning across varying operating conditions, thereby limiting the practical implementation of DPSC. To solve this problem, many scholars have investigated the coupling issue between dual time-scale methods in the context of predictive speed control applications within cascade speed regulation systems. Ref. [160] proposed the development of predictive speed and current control to achieve both fast and accurate predictive control. In [161], a virtual-torque-based linear estimation method to enhance the accuracy of slowly sampled models used for rapid torque prediction is proposed. However, conventional observers designed based on the error between actual and measured values demonstrate pronounced time lag when subjected to step signal disturbances. To mitigate this issue, paper [70] developed a Predictive Error Model Observer (PEMO) that proactively estimates error trends during model mismatch conditions. Although this approach exhibits faster convergence and reduced time delay characteristics, it consequently increases the predictive results' sensitivity to system noise.

To address the limitations of the aforementioned methods, paper [162] proposed a robust cascaded deadbeat predictive speed control scheme incorporating both speed and disturbance observers. The developed cascaded DPSC system employs a speed observer to mitigate inherent time delay issues. Furthermore, to enhance system robustness against parameter variations and load torque disturbances, a composite torque disturbance observer was introduced, combining a generalized proportional–integral observer with a sliding mode observer, demonstrating superior control performance; however, this method will increase the current disturbance on the d-axis. For enhancing the load rejection capability and parameter robustness of PMSM DPSC systems, paper [163] proposed a low-gain disturbance observer with inertia identification. Through precise inertia identification, the observer bandwidth is significantly reduced, which effectively suppresses measurement noise interference while satisfying PMSM stiffness constraints. To address the slow convergence issue caused by low bandwidth, an RLS-based prediction error model (PEM) was employed to predict speed variation trends. This approach enables fast observer convergence during transient periods while maintaining the advantages of non-overshooting characteristics and strong noise rejection. Table 4 lists the performance of the proposed methods in [162,163] and conventional methods.

Table 4. Performance summary of various methods.

Methods	PI-DPCC	DPCC-ESO	Method in [Robust Ca]	Method in [Zuihou]
Speed tracking (setting/rising time)	140 ms/160 ms	110 ms/250 ms	112 ms/125 ms	130 ms/144 ms
Load disturbance rejection (speed drop/recovery time)	40 rpm/400 ms	32 rpm/230 ms	28 rpm/200 ms	26 rpm/228 ms
Steady-state Performance (speed ripples/current ripples)	2 rpm/0.1 A	2 rpm/0.3 A	2 rpm/0.5 A	1.4 rpm/0.3 A

4. Future Directions

Through the introduction and summary of the existing literature, the deadbeat predictive control method has garnered significant attention and undergone extensive research

in PMSM applications. However, as industrial and other application scenarios demand increasingly stringent control performance requirements, deadbeat predictive control must adapt to more challenging operating conditions. The current combination of existing methods with DPC is shown in Table 5 and current research directions primarily focus on the following aspects.

Table 5. Combination of existing methods with DPC.

Methods	Features of Application
Parameter identification + DPCC	Parameter mismatch
Observers + DPCC	Disturbances and complex working conditions
Error compensation + DPCC	Parameter mismatch or complex working conditions
Delay compensation + DPC	Low computing power equipment and high-precision instrument
Observers + DPSC	Disturbances and quick response time conditions

4.1. Enhancement of Robustness Against Model Uncertainties

Although deadbeat predictive control achieves exceptional dynamic performance in permanent magnet synchronous motor (PMSM) drives, its effectiveness remains compromised by sensitivity to parameter variations [57,85,164]. It also exhibits degraded steady-state performance under inductance and flux linkage mismatches, manifesting as pronounced torque ripple. Conventional robustness enhancement strategies relying on disturbance observers and online parameter identification increase system complexity while requiring meticulous observer bandwidth tuning [129,133,165–167]. Cross-saturation effects further diminish estimation accuracy during field-weakening operation. Consequently, developing model-agnostic control frameworks with inherent robustness constitutes a critical research direction.

4.2. Advanced Delay Compensation Techniques

Multi-step prediction strategies have been adopted to mitigate computational and sampling delays. However, extending prediction horizons substantially increases computational demands. While prediction horizon reduction techniques alleviate computational burdens, they depend critically on precise position sensing and accurate mechanical parameters. Practical implementations face significant challenges from position signal interference and time-varying motor parameters [149,168–170]. Notably, uncompensated delays persist during high-speed operation, causing observable current distortion. Therefore, adaptive delay compensation mechanisms with self-calibrating capabilities warrant focused investigation.

4.3. Optimization and Improvement in Comprehensive Disturbance Observer

The conventional single observer structure has inherent limitations: the sliding mode observer (SMO) provides robust anti-interference but causes high-frequency jitter [115,122], while the generalized proportional–integral (GPI) observer demonstrates superior anti-noise capability at the cost of a slower convergence speed [151]. Extended State tube porcelain (ESO) has outstanding convergence speed and anti-noise ability, but it also heavily relies on the system's bandwidth [94,112,114]. Although the cascade observer structure improves the suppression of compound disturbances, it introduces complex parameter tuning requirements. Under time-varying conditions, the interaction of observers may cause torque estimation bias [72,171–176]. Therefore, cascaded observers with anti-interference ability and adaptive adjustment have broad research prospects.

4.4. Multi-Objective Optimization Frameworks

The existing DPC implementation efforts simultaneously optimize the competitive objective: parameter mismatch can lead to a decline in control performance, and although the parameter identification method can reduce the degradation of control performance, it will also increase the response time [136,137]. Suppressing torque ripple requires a high switching frequency, but it will intensify inverter losses, while minimizing switching losses will increase current harmonics. How to optimize multiple targets that affect control performance simultaneously without affecting other parameters is the current research focus [118–122].

4.5. Computationally Efficient Implementations

Although deadbeat predictive control (DPC) delivers superior dynamic performance in PMSM drives, its real-time implementation faces significant computational bottlenecks. The strict execution time requirement necessitates highly optimized algorithms. Conventional approaches involve simplified voltage vector selection and reduced-order models, but these compromise current tracking accuracy during transients. Lookup table methods accelerate duty cycle calculation at the expense of substantial memory allocation. Hardware solutions like FPGA-based parallelization improve cycle efficiency but increase design complexity and cost. Consequently, developing adaptive algorithms that dynamically balance computational load and control precision remains critical for high-speed applications.

5. Conclusions

This paper summarizes and reviews the research progress of permanent magnet synchronous motors. Through the introduction and classification of the basic principles of DPC, the problems existing in the application of DPC in the current loop and speed loop were analyzed, and the existing optimization schemes were reviewed. The robustness of DPC based on discrete models is most worthy of consideration and optimization. In addition, parameter identification, delay compensation, harmonic suppression, and reduction in computational load have also been widely studied. However, there is still room for research and development on the related issues discussed above. For instance, most harmonic suppression methods will reduce the robustness of the system. Therefore, how to suppress harmonics without affecting system performance is currently the focus of research. With the rapid development of chip processors, the application of predictive control is also increasing. The combination of DPC with other existing control methods is also a direction worthy of research and exploration in the future.

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References

1. Li, T.; Sun, X.; Lei, G.; Yang, Z.; Guo, Y.; Zhu, J. Finite-control-set model predictive control of permanent magnet synchronous motor drive systems—An overview. *IEEE/CAA J. Autom. Sin.* **2022**, *9*, 2087–2105. [CrossRef]
2. Wu, G.; Huang, S.; Wu, Q.; Rong, F.; Zhang, C.; Liao, W. Robust predictive torque control of N*3-phase PMSM for high-power traction application. *IEEE Trans. Power Electron.* **2020**, *35*, 10799–10809. [CrossRef]
3. Sun, X.; Xu, Z.; Pan, M.; Sun, C.; Lei, G. Medium-to-High Speed Range Sensorless Control of SRM Drives Based on Unaligned Rotor Position Estimation. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]
4. Lei, G.; Zhu, J.G.; Guo, Y.G.; Liu, C.C.; Ma, B. A review of design optimization methods for electrical machines. *Energies* **2017**, *10*, 1962. [CrossRef]
5. Sun, X.; Xu, Z.; Pan, M.; Cai, Y.; Jin, Z.; Tian, X. Driving-Style-Oriented Energy Management Strategy Based on Fuzzy Logic for Plug-in Hybrid Electric Buses. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]
6. Yue, R.; Yao, M.; Zhang, T.; Shi, J.; Zhou, J.; Hu, J. Design and experiment of dual-row seedling Pick-Up device for High-Speed automatic transplanting machine. *Agriculture* **2024**, *14*, 942. [CrossRef]
7. Niu, S.X.; Luo, Y.X.; Fu, W.N.; Zhang, X.D. An indirect reference vector-based model predictive control for a three-phase PMSM motor. *IEEE Access* **2020**, *8*, 29435–29445. [CrossRef]
8. Jin, Z.; Sun, X.; Cai, Y.; Yao, M.; Tian, X. Simultaneous-Level Energy Management Optimization Considering Specific Urban Driving Cycles for PHEBs. *IEEE/ASME Trans. Mechatron.* 2024; *early access*. [CrossRef]
9. Sun, X.; Xu, N.; Yao, M.; Cai, F.; Wu, M. Efficient feedback linearization control for an IPMSM of EVs based on improved firefly algorithm. *ISA Trans.* **2023**, *134*, 431–441. [CrossRef]
10. Shao, L.; Hua, W.; Dai, N.; Tong, M.; Cheng, M. Mathematical Modeling of a 12-Phase Flux-Switching Permanent-Magnet Machine for Wind Power Generation. *IEEE Trans. Ind. Electron.* **2016**, *63*, 504–516. [CrossRef]
11. Ding, X.F.; Guo, H.; Xiong, R.; Chen, F.D.; Zhang, D.H.; Gerada, C. A new strategy of efficiency enhancement for traction systems in electric vehicles. *Appl. Energy* **2017**, *205*, 880–891. [CrossRef]
12. Hao, S.; Tang, Z.; Guo, S.; Ding, Z.; Su, Z. Model and method of fault signal diagnosis for blockage and slippage of rice threshing drum. *Agriculture* **2022**, *12*, 1968. [CrossRef]
13. Li, Y.; Liu, Y.; Ji, K.; Zhu, R. A Fault Diagnosis Method for a Differential Inverse Gearbox of a Crawler Combine Harvester Based on Order Analysis. *Agriculture* **2022**, *12*, 1300. [CrossRef]
14. Trancho, E.; Ibarra, E.; Arias, A.; Kortabarria, I.; Prieto, P.; De Alegría, I.M.; Andreu, J.; López, I. Sensorless control strategy for light-duty EVs and efficiency loss evaluation of high frequency injection under standardized urban driving cycles. *Appl. Energy* **2018**, *224*, 647–658. [CrossRef]
15. Shi, Z.; Sun, X.; Yang, Z.; Cai, Y.; Lei, G.; Zhu, J.; Lee, C.H.T. Design optimization of a spoke type axial-flux PM machine for in-wheel drive operation. *IEEE Trans. Transp. Electrif.* **2024**, *10*, 3770–3781. [CrossRef]
16. Li, T.; Sun, X.; Su, Z.; Zha, X.; Dianov, A.; Prakht, V.; Demidova, G.; Ma, J. Model-Free Predictive Control for Harmonic Suppression of PMSMs Based on Adaptive Resonant Controller. *IEEE Trans. Energy Convers.* 2025; *early access*. [CrossRef]
17. Wang, S. Nonlinear uncertainty estimator based robust control for PMSM servo mechanisms with prescribed performance. *IEEE Trans. Transp. Electrific.* **2023**, *9*, 2535–2543. [CrossRef]
18. Zhou, Y.; Chen, G. Predictive DTC strategy with fault-tolerant function for six-phase and three-phase PMSM series-connected drive system. *IEEE Trans. Ind. Electron.* **2018**, *65*, 9101–9112. [CrossRef]
19. Yang, S.; Zhai, C.; Gao, Y.; Dou, H.; Zhao, X.; He, Y.; Wang, X. Planting uniformity performance of motor-driven maize precision seeding systems. *Int. J. Agric. Biol. Eng.* **2022**, *15*, 101–108. [CrossRef]
20. Yu, F.; Zhang, W.; Shen, Y.; Mao, J. A Nine-Phase Permanent Magnet Electric-Drive-Reconstructed Onboard Charger for Electric Vehicle. *IEEE Trans. Energy Convers.* **2018**, *33*, 2091–2101. [CrossRef]
21. Hao, Z.; Zhu, H.; Cheng, Y.; Huang, L. Speed Control of Bearingless Permanent Magnet Synchronous Motor Based on Flux Strengthening and Voltage Regulation. *IEEE Access* **2018**, *6*, 72392–72401. [CrossRef]
22. Hu, Y.; Li, Y.; Ma, X.; Li, X.; Huang, S. Flux-weakening control of dual three-phase PMSM based on vector space decomposition control. *IEEE Trans. Power Electron.* **2021**, *36*, 8428–8438. [CrossRef]
23. Sun, X.; Zhang, S.; An, X.; Su, Z.; Li, F. A robust DPCC and inductance identification method for SPMSM based on incremental model and current error compensation. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]
24. Gu, M.; Wang, Z.; Xia, W.; Cheng, M. Analysis and Modeling of Switching Surges in PMSM Drives Due to Eddy-Current Reaction. *IEEE Trans. Ind. Electron.* **2025**, *72*, 5681–5692. [CrossRef]
25. Yao, H.; Yan, Y.; Shi, T.; Zhang, G.; Wang, Z.; Xia, C. A novel SVPWM scheme for field-oriented vector-controlled PMSM drive system fed by cascaded H-bridge inverter. *IEEE Trans. Power Electron.* **2021**, *36*, 8988–9000. [CrossRef]
26. Sun, X.; Zhang, H.; Lin, X.; Su, Z.; Yao, M. Speed sensorless control based on model -free predictive controller for dual three-phase PMSMs. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]

27. Han, J.; Wang, F. Design and testing of a small orchard tractor driven by a power battery. *Eng. Agric.* **2023**, *43*, e20220195. [CrossRef]
28. Pang, Y.; Li, H.; Tang, P.; Chen, C. Irrigation scheduling of pressurized irrigation networks for minimizing energy consumption. *Irrig. Drain.* **2023**, *72*, 268–283. [CrossRef]
29. Ping, Z.; Ma, Q.; Wang, T.; Huang, Y.; Lu, J.-G. Speed tracking control of permanent magnet synchronous motor by a novel two-step internal model control approach. *Int. J. Control Autom. Syst.* **2018**, *16*, 2754–2762. [CrossRef]
30. Wang, Z.; Chen, J.; Cheng, M.; Chau, K.T. Field-oriented control and direct torque control for paralleled VSIs fed PMSM drives with variable switching frequencies. *IEEE Trans. Power Electron.* **2016**, *31*, 2417–2428. [CrossRef]
31. Lee, K. Fault-Tolerant Control for One-Phase Current Sensor Fault in Open-End Winding PMSM fed by Dual Inverter With a Common DC Bus. *IEEE Trans. Power Electron.* **2025**, *40*, 124–128. [CrossRef]
32. Rubio, F.; Pereda, J.; Mora, A.; Rojas, F. Optimal Switching Sequence MPC of a Three- Port-Converter for Variable-Speed PMSM With Hybrid Energy Storage. *IEEE Trans. Power Electron.* **2025**, *40*, 5069–5079. [CrossRef]
33. Zhang, Q.F.; Guo, H.H.; Guo, C.; Liu, Y.C.; Wang, D.; Lu, K.Y.; Zhang, Z.R.; Zhuang, X.Z.; Chen, D.Z. An adaptive proportionalintegral-resonant controller for speed ripple suppression of PMSM drive due to current measurement error. *Int. J. Electr. Power Energy Syst.* **2021**, *129*, 106866. [CrossRef]
34. Casadei, D.; Profumo, F.; Serra, G.; Tani, A. FOC and DTC: Two viable schemes for induction motors torque control. *IEEE Trans. Power Electron.* **2002**, *17*, 779–787. [CrossRef]
35. Xiang, Z.; Wei, J.; Zhu, X. Torque ripple suppression of a PM vernier machine from Perspective of time and space harmonic magnetic field. *IEEE Trans. Ind. Electron.* **2024**, *71*, 10150–10161. [CrossRef]
36. Yu, Y.; Hao, S.; Guo, S.; Tang, Z.; Chen, S. Motor torque distribution strategy for different tillage modes of agricultural electric tractors. *Agriculture* **2022**, *12*, 1373. [CrossRef]
37. Shinohara, A.; Inoue, Y.; Morimoto, S.; Sanada, M. Maximum Torque Per Ampere Control in Stator Flux Linkage Synchronous Frame for DTC-Based PMSM Drives Without Using q-Axis Inductance. *IEEE Trans. Ind. Appl.* **2017**, *53*, 3663–3671. [CrossRef]
38. Ding, L.; Li, Y.W.; Zargari, N.R. Discrete-time SMO sensorless control of current source converter-fed PMSM drives with low switching frequency. *IEEE Trans. Ind. Electron.* **2021**, *68*, 2120–2129. [CrossRef]
39. Wu, L.; Liu, J.; Vazquez, S.; Mazumder, S.K. Sliding mode control in power converters and drives: A review. *IEEE/CAA J. Autom. Sin.* **2022**, *9*, 392–406. [CrossRef]
40. Lian, S.; Meng, W.; Lin, Z.; Shao, K.; Zheng, J.; Li, H.; Lu, R. Adaptive attitude control of a quadrotor using fast nonsingular terminal sliding mode. *IEEE Trans. Ind. Electron.* **2022**, *69*, 1597–1607. [CrossRef]
41. Xu, W.; Junejo, A.K.; Liu, Y.; Hussien, M.G.; Zhu, J. An efficient antidisturbance sliding-mode speed control method for PMSM drive systems. *IEEE Trans. Power Electron.* **2021**, *36*, 6879–6891. [CrossRef]
42. Wu, S.; Su, X.; Wang, K. Time-dependent global nonsingular fixed-time terminal sliding mode control-based speed tracking of permanent magnet synchronous motor. *IEEE Access* **2020**, *8*, 186408–186420. [CrossRef]
43. Gou, L.; Wang, C.; Zhou, M.; You, X. Integral sliding mode control for starting speed sensorless controlled induction motor in the rotating condition. *IEEE Trans. Power Electron.* **2020**, *35*, 4105–4116. [CrossRef]
44. Xu, P.L.; Zhu, Z.Q. Novel Carrier Signal Injection Method Using Zero-Sequence Voltage for Sensorless Control of PMSM Drives. *IEEE Trans. Ind. Electron.* **2016**, *63*, 2053–2061. [CrossRef]
45. Verrelli, C.M.; Bifaretti, S.; Carfagna, E.; Lidozzi, A.; Solero, L.; Crescimbini, F.; Di Benedetto, M. Speed Sensor Fault Tolerant PMSM Machines: From Position-Sensorless to Sensorless Control. *IEEE Trans. Ind. Appl.* **2019**, *55*, 3946–3954. [CrossRef]
46. Luo, B.; Yang, X.; Zhou, Y. Model-Free Predictive Current Control of Permanent Magnet Synchronous Motor Based on Estimation of Current Variations. *IEEE Trans. Ind. Electron.* **2024**, *71*, 8395–8405. [CrossRef]
47. Sun, X.; Li, T.; Zhu, Z.; Lei, G.; Guo, Y.; Zhu, J. Speed sensorless model predictive current control based on finite position set for PMSHM Drives. *IEEE Trans. Transport. Electrific.* **2021**, *7*, 2743–2752. [CrossRef]
48. Sun, X.; Cai, F.; Yang, Z.; Tian, X. Finite position control of interior permanent magnet synchronous motors at low speed. *IEEE Trans. Power Electron.* **2022**, *37*, 7729–7738. [CrossRef]
49. Novak, Z. Confidence Weighted Learning Entropy for Fault-Tolerant Control of a PMSM With a High-Resolution Hall Encoder. *IEEE Trans. Ind. Electron.* **2024**, *71*, 5176–5186. [CrossRef]
50. Zhang, H.; Ji, W.; Xu, B.; Yu, X. Optimizing contact force on an apple picking robot end-effector. *Agriculture* **2024**, *14*, 996. [CrossRef]
51. Shi, J.; Hu, J.; Li, J.; Liu, W.; Yue, R.; Zhang, T.; Yao, M. Design and experiment of planting mechanism of automatic transplanter for densely planted vegetables. *Agriculture* **2024**, *14*, 1357. [CrossRef]
52. Jin, L.; Mao, Y.; Wang, X.; Shi, P.; Lu, L.; Wang, Z. Optimization-Based Maximum-Torque Fault-Tolerant Control of Dual Three-Phase PMSM Drives Under Open-Phase Fault. *IEEE Trans. Power Electron.* **2023**, *38*, 3653–3663. [CrossRef]
53. Wu, L.; Guo, Y.; Huang, X.; Fang, Y.; Liu, J. Harmonic Torque Suppression Methods for Single-Phase Open-Circuit Fault-Tolerant Operation of PMSM Considering Third Harmonic BEMF. *IEEE Trans. Power Electron.* **2021**, *36*, 1116–1129. [CrossRef]

54. He, Y.; Zhu, Q.; Fu, W.; Luo, C.; Cong, Y.; Qin, W.; Meng, Z.; Chen, L.; Zhao, C.; Wu, G. Design and experiment of a control system for sweet potato seedling-feeding and planting device based on a pre-treatment seedling belt. *J. Agric. Eng.* **2022**, *53*, 1261. [CrossRef]
55. Zhang, Y.; Chen, X.; Wang, X.; Zhang, T.; Wang, Z.; Mao, Y. Online Current Optimization Based Fault-Tolerant Control of Standard PMSM Drives. *IEEE Trans. Ind. Electron.* **2024**, *71*, 15521–15531. [CrossRef]
56. Liu, S.; Liu, C.; Huang, Y.; Zhao, H. Model predictive two-target current control for OW-PMSM. *IEEE Trans. Power Electron.* **2021**, *36*, 3224–3235. [CrossRef]
57. Li, X.; Tian, W.; Gao, X.; Yang, Q.; Kennel, R. A generalized observer-based robust predictive current control strategy for PMSM drive system. *IEEE Trans. Ind. Electron.* **2022**, *69*, 1322–1332. [CrossRef]
58. Su, Z.; Sun, X.; Lei, G.; Yao, M. Model-free predictive current control for dual three-phase PMSM drives with an optimal modulation pattern. *IEEE Trans. Ind. Electron.* **2024**, *71*, 10140–10149. [CrossRef]
59. Sun, X.; Huang, Z.; Yang, Z.; Lei, G.; Li, T. Improved Model-Free Predictive Current Control for Suppressing Inverter Nonlinearity and Parametric Time-Varying of PMSM Drive Systems. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]
60. Lian, C.; Guo, L.; Xu, G.; Xiao, F.; Wang, R.; Gao, S.; Liu, J.; Zhang, W. Current Harmonic Minimum Pulse Width Modulation for Dual Three-Phase PMSM System With 3L-NPC Inverter and Output Filter. *IEEE Trans. Power Electron.* **2025**, *40*, 2847–2859. [CrossRef]
61. Zhou, Z.; Xia, C.; Yan, Y.; Wang, Z.; Shi, T. Torque Ripple Minimization of Predictive Torque Control for PMSM With Extended Control Set. *IEEE Trans. Ind. Electron.* **2017**, *64*, 6930–6939. [CrossRef]
62. Wang, X.; Fang, X.; Lin, S.; Lin, F.; Yang, Z. Predictive common-mode voltage suppression method based on current ripple for permanent magnet synchronous motors. *IEEE J. Emerg. Sel. Top. Power Electron.* **2019**, *7*, 946–955. [CrossRef]
63. Li, T.; Sun, X.; Yao, M.; Guo, D.; Sun, Y. Improved Finite Control Set Model Predictive Current Control for Permanent Magnet Synchronous Motor with Sliding Mode Observer. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 699–710. [CrossRef]
64. Chen, Y.; Wang, X.; Xiao, D.; Ma, D.; Yang, X.; Wang, Z. Model Predictive Control of Five-Level Open-End Winding PMSM Drives. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 10115–10124. [CrossRef]
65. Deng, J.; Ni, L.; Bai, X.; Jiang, H.; Xu, L. Simultaneous analysis of mildew degree and aflatoxin B1 of wheat by a multi-task deep learning strategy based on microwave detection technology. *LWT* **2023**, *184*, 115047. [CrossRef]
66. Yao, C.; Ma, G.; Sun, Z.; Luo, J.; Ren, G.; Xu, S. Weighting Factors Optimization for FCS-MPC in PMSM Drives Using Aggregated Residual Network. *IEEE Trans. Power Electron.* **2024**, *39*, 1292–1307. [CrossRef]
67. Xu, S.; Yao, C.; Ren, G.; Sun, Z.; Wu, S.; Ma, G. Weighting Factors Autotuning of FCS-MPC for Hybrid ANPC Inverter in PMSM Drives Based on Deep Residual Networks. *IEEE Trans. Power Electron.* **2024**, *39*, 16540–16552. [CrossRef]
68. Jiang, X.; Yang, Y.; Fan, M.; Ji, A.; Xiao, Y.; Zhang, X.; Zhang, W.; Garcia, C.; Vazquez, S.; Rodriguez, J. An Improved Implicit Model Predictive Current Control with Continuous Control Set for PMSM Drives. *IEEE Trans. Power Electron.* **2022**, *8*, 2444–2455. [CrossRef]
69. Liu, Z.; Huang, X.; Hu, Q.; Li, Z.; Jiang, Z.; Yu, Y.; Chen, Z. A Modified Deadbeat Predictive Current Control for Improving Dynamic Performance of PMSM. *IEEE Trans. Power Electron.* **2022**, *37*, 14173–14185. [CrossRef]
70. Ke, D.; Wang, F.; Yu, X.; Davari, S.A.; Kennel, R. Predictive Error Model-Based Enhanced Observer for PMSM Deadbeat Control Systems. *IEEE Trans. Ind. Electron.* **2024**, *71*, 2242–2252. [CrossRef]
71. Chen, Y.; Liu, C.; Liu, S.; Song, Z. A New Cascaded Adaptive Deadbeat Control Method for PMSM Drive. *IEEE Trans. Ind. Electron.* **2023**, *70*, 3384–3393. [CrossRef]
72. Wang, W.; Liu, C.; Song, Z.; Dong, Z. Harmonic Current Suppression for Dual Three-Phase PMSM Based on Deadbeat Control and Disturbance Observer. *IEEE Trans. Ind. Electron.* **2023**, *70*, 3482–3492. [CrossRef]
73. Li, X.; Yang, Y.; Sun, J.; Xiao, Y.; Fan, M.; Ni, K.; Hu, J.; Wen, H.; Yang, H.; Rodriguez, J. Multiple-Voltage-Vector Model-Free Predictive Deadbeat Control with Updated Reference Voltage Vector for PMSM Drive. *IEEE Trans. Power Electron.* **2025**, *40*, 6492–6505. [CrossRef]
74. Chen, K.; Li, T.; Yan, T.; Xie, F.; Feng, Q.; Zhu, Q.; Zhao, C. A soft gripper design for apple harvesting with force feedback and fruit slip detection. *Agriculture* **2022**, *12*, 1802. [CrossRef]
75. Huan, J.; Cao, W.; Liu, X. Technical note: A dissolved oxygen prediction method based on K-means clustering and the ELM neural network: A case study of the Changdang Lake. *Appl. Eng. Agric.* **2017**, *33*, 461–469. [CrossRef]
76. Nguyen, T.T.; Tran, H.N.; Nguyen, T.H.; Jeon, J.W. Recurrent Neural Network-Based Robust Adaptive Model Predictive Speed Control for PMSM With Parameter Mismatch. *IEEE Trans. Ind. Electron.* **2023**, *70*, 6219–6228. [CrossRef]
77. Duan, J.; Cao, P.; Shu, T.; Zhou, B.; Xue, L.; Yang, L. Effects of Cellulosic Carbon Addition on Nitrogen Removal from Simulated Dry Land Drainage, and Its Environmental Effects. *Agronomy* **2023**, *13*, 3044. [CrossRef]
78. Lin, X.; Liu, J.; Liu, Z.; Gao, Y.; Peretti, L.; Wu, L. Model-Free Current Predictive Control for PMSMs With Ultra local Model Employing Fixed-Time Observer and Extremum-Seeking Method. *IEEE Trans. Power Electron.* **2025**, *40*, 10682–10693. [CrossRef]

79. Xu, Y.; Zhang, Y.; Gao, Q.; Yin, Z. A Wide-Speed Range Three-Vector-Based Model Predictive Flux Control with Improved Predictive Model for PMSM Drives. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 2689–2700. [CrossRef]
80. Wu, X.; Kang, J.; Yang, M.; Wu, T.; Huang, S. Model-Free Deadbeat Predictive Current Control for SPMSM Based on Adaptive Gain Extended State Observer. *IEEE Trans. Transp. Electrific.* **2025**, *11*, 536–543. [CrossRef]
81. Cao, H.; Deng, Y.; Liu, J.; Zuo, Y.; Liu, X.; Wang, H.; Lee, C.H.T. Improved Deadbeat Predictive Current Control of PMSM Drives with Repetitive Control-Based Disturbance Correction Observer. *IEEE Trans. Power Electron.* **2025**, *40*, 801–812. [CrossRef]
82. Wang, L.; Zhang, S.; Zhang, C.; Zhou, Y. An Improved Deadbeat Predictive Current Control Based on Parameter Identification for PMSM. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 2740–2753. [CrossRef]
83. Xu, Q.; Fang, S. Robust Deadbeat Predictive Current Sensorless Control for Permanent Magnet Synchronous Motor. *IEEE Trans. Power Electron.* **2024**, *39*, 11441–11454. [CrossRef]
84. Jin, Z.; Sun, X.; Cai, Y.; Tian, X. Robust collaborative optimization design of plug-in hybrid electric bus based on 6 Sigma theory. *IEEE Trans. Transport. Electrific.* **2024**, *10*, 10253–10261. [CrossRef]
85. Hang, J.; Zhang, J.; Ding, S.; Huang, Y.; Wang, Q. A Model-Based Strategy with Robust Parameter Mismatch for Online HRC Diagnosis and Location in PMSM Drive System. *IEEE Trans. Power Electron.* **2020**, *35*, 10917–10929. [CrossRef]
86. Zhou, J.; Yang, J.; Miao, H.; Fan, X.; Wang, Y.; Song, D. Shift-MRF Based Current Harmonics Compensation Decoupling Strategy in PMSM Considering Sampling Delay and Parameter Mismatch. *IEEE Trans. Ind. Electron.* **2025**, *72*, 5625–5635. [CrossRef]
87. Layek, K.; Nair, S.V.; Krishnan, P.H.; Hatua, K. Voltage Angle-Based Torque Control of Dual Three-Phase Interior Permanent Magnet Synchronous Motor Considering Mismatch in Winding Parameters. *IEEE Trans. Ind. Electron.* **2024**, *71*, 10119–10129. [CrossRef]
88. Zhu, Q.; Zhu, Z.; Zhang, H.; Gao, Y.; Chen, L. Design of an electronically controlled fertilization system for an air-assisted side-deep fertilization machine. *Agriculture* **2023**, *13*, 2210. [CrossRef]
89. Sun, X.; Lin, X.; Guo, D.; Lei, G.; Yao, M. Improved deadbeat predictive current control with extended state observer for dual three-phase PMSMs. *IEEE Trans. Power Electron.* **2024**, *39*, 6769–6782. [CrossRef]
90. Dong, X.; Mao, J.; Yan, Y.; Zhang, C.; Yang, J. Generalized Dynamic Predictive Control for Nonlinear Systems Subject to Mismatched Disturbances with Application to PMSM Drives. *IEEE Trans. Ind. Electron.* **2024**, *71*, 954–964. [CrossRef]
91. Li, X.; Zhang, S.; Zhang, C.; Zhou, Y.; Zhang, C. An Improved Deadbeat Predictive Current Control Scheme for Open-Winding Permanent Magnet Synchronous Motors Drives with Disturbance Observer. *IEEE Trans. Power Electron.* **2021**, *36*, 4622–4632. [CrossRef]
92. Yang, F.; Zhao, X.; Jin, H.; Wang, X.; Liu, X. Improved Deadbeat Predictive Current Control with Embedded Resonant Polynomial and Disturbance Observer for PMSM Current Distortion Rejection. *IEEE J. Emerg. Sel. Top. Power Electron.* **2024**, *12*, 1934–1945. [CrossRef]
93. Dai, S.; Wang, J.; Sun, Z.; Chong, E. Model Inaccuracy Analysis and Compensation of Stationary Frame-Based Deadbeat Predictive Current Control for High-Speed PMSM Drives. *IEEE Trans. Transp. Electrific.* **2022**, *8*, 2654–2666. [CrossRef]
94. Zhang, Q.; Fan, Y.; Mao, C. A Gain Design Method for a Linear Extended State Observers to Improve Robustness of Deadbeat Control. *IEEE Trans. Energy Convers.* **2020**, *35*, 2231–2239. [CrossRef]
95. Rovere, L.; Formentini, A.; Zanchetta, P. FPGA Implementation of a Novel Oversampling Deadbeat Controller for PMSM Drives. *IEEE Trans. Ind. Electron.* **2019**, *66*, 3731–3741. [CrossRef]
96. Sun, J.; Nirere, A.; Dusabe, K.D.; Yuhao, Z.; Adrien, G. Rapid and nondestructive watermelon (*Citrullus lanatus*) seed viability detection based on visible near-infrared hyperspectral imaging technology and machine learning algorithms. *J. Food Sci.* **2024**, *89*, 4403–4418. [CrossRef]
97. Huang, X.; Hu, Q.; Liu, Z.; Li, W.; Yang, G.; Li, Z. A Robust Deadbeat Predictive Current Control Method for IPMSM. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 4722–4733. [CrossRef]
98. Jiang, Y.; Xu, W.; Mu, C.; Liu, Y. Improved Deadbeat Predictive Current Control Combined Sliding Mode Strategy for PMSM Drive System. *IEEE Trans. Veh. Technol.* **2018**, *67*, 251–263. [CrossRef]
99. Liu, Z.-H.; Wei, H.-L.; Li, X.-H.; Liu, K.; Zhong, Q.-C. Global Identification of Electrical and Mechanical Parameters in PMSM Drive Based on Dynamic Self-Learning PSO. *IEEE Trans. Power Electron.* **2018**, *33*, 10858–10871. [CrossRef]
100. Inoue, Y.; Kawaguchi, Y.; Morimoto, S.; Sanada, M. Performance Improvement of Sensorless IPMSM Drives in a Low-Speed Region Using Online Parameter Identification. *IEEE Trans. Ind. Appl.* **2011**, *47*, 798–804. [CrossRef]
101. Xie, C.; Zhang, S.; Li, X.; Zhou, Y.; Dong, Y. Parameter Identification for SPMSM With Deadbeat Predictive Current Control Using Online PSO. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 4055–4064. [CrossRef]
102. Xiang, D.; Yang, J.; Hao, Y.; Xu, G. Parallel-Cascaded Parameter Identification Scheme for PMSM-Driven Servo Systems During Self-Commission. *IEEE Trans. Ind. Electron.* **2025**, *72*, 1914–1924. [CrossRef]
103. Zhang, Y.; Zhou, M.; Zhang, C.; Shen, A.; Bing, L. Identification of PMSM Parameters with Time-Error Compensated Based on Contractile Factor Antipredator PSO. *IEEE Trans. Transp. Electrific.* **2024**, *10*, 4006–4017. [CrossRef]

104. Wang, Y.; Fang, S.; Hu, J.; Huang, D. Multiscenarios Parameter Optimization Method for Active Disturbance Rejection Control of PMSM Based on Deep Reinforcement Learning. *IEEE Trans. Ind. Electron.* **2023**, *70*, 10957–10968. [CrossRef]
105. Liu, S.; Tang, Z.; Zhang, B.; Liang, Y.; Gu, X. Ergonomic design of cab structure for wheeled combine harvester. *Eng. Agric.* **2022**, *42*, e20220023. [CrossRef]
106. Tang, Z.; Wang, H.; Jing, T.; Liu, S.; Shen, C. Centroid swinging property of a machine body undergoing longitudinal excitation in a straw compression baler. *Eng. Agric.* **2024**, *44*, e20230109. [CrossRef]
107. Zhang, X.; Wang, P.; Mao, H.; Gao, H.; Li, Q. Detection of the nutritional status of phosphorus in lettuce using THz time-domain spectroscopy. *Eng. Agric.* **2021**, *41*, 599–608. [CrossRef]
108. Zhang, Y.; Li, S.; Yi, W.; Yang, Y.; Cao, K.; Luo, B. A Robust Model Predictive Current Closed-Loop Control with Parameter Estimation Strategy Based on Immune Chaotic Antipredator PSO for PMSM. *IEEE Trans. Power Electron.* **2025**, *40*, 1836–1848. [CrossRef]
109. Liu, Z.-H.; Nie, J.; Wei, H.-L.; Chen, L.; Wu, F.-M.; Lv, M.-Y. Second-Order ESO-Based Current Sensor Fault-Tolerant Strategy for Sensorless Control of PMSM With B-Phase Current. *IEEE/ASME Trans. Mechatron.* **2022**, *27*, 5427–5438. [CrossRef]
110. Su, Z.; Sun, X.; Lei, G.; Yao, M. Improved model-free predictive current control for SPMSM drives with adaptive prediction horizon strategy. *IEEE Trans. Ind. Electron.* **2025**, *72*, 5705–5715. [CrossRef]
111. Sun, X.; Su, Z.; Lei, G.; Cai, Y.; Yao, M. Adaptive model-free predictive current control for SPMSM drives with optimal virtual vector modulation. *IEEE/ASME Trans. Mechatron.* **2024**, *29*, 2569–2578. [CrossRef]
112. Xiong, J.; Fu, X. Extended Two-State Observer-Based Speed Control for PMSM With Uncertainties of Control Input Gain and Lumped Disturbance. *IEEE Trans. Ind. Electron.* **2024**, *71*, 6172–6182. [CrossRef]
113. Hou, Q.; Xu, S.; Zuo, Y.; Wang, H.; Sun, J.; Lee, C.H.T.; Ding, S. Enhanced Active Disturbance Rejection Control with Measurement Noise Suppression for PMSM Drives via Augmented Nonlinear Extended State Observer. *IEEE Trans. Energy Convers.* **2024**, *39*, 287–299. [CrossRef]
114. Song, J.; Wang, Y.-K.; Niu, Y.; Lam, H.-K.; He, S.; Liu, H. Periodic Event-Triggered Terminal Sliding Mode Speed Control for Networked PMSM System: A GA-Optimized Extended State Observer Approach. *IEEE/ASME Trans. Mechatron.* **2022**, *27*, 4153–4164. [CrossRef]
115. Wang, B.; Wang, T.; Yu, Y.; Xu, D. Second-order terminal sliding mode speed controller for induction motor drives with nonlinear control gain. *IEEE Trans. Ind. Electron.* **2023**, *70*, 10923–10934. [CrossRef]
116. Sun, Y.; Yang, M.; Wang, B.; Chen, Y.; Xu, D. Precise position control based on resonant controller and second-order sliding mode observer for PMSM-driven feed servo system. *IEEE Trans. Transport. Electrific.* **2023**, *9*, 196–209. [CrossRef]
117. Sun, X.; Lin, X.; Han, S.; Wang, Y. An improved continuous-control-set model predictive control for dual three-phase PMSMs based on extended state observer. *IEEE Trans. Ind. Electron.* **2025**, *72*, 3449–3459. [CrossRef]
118. Liang, Z.; Li, Y.; Xu, L.; Zhao, Z. Sensor for monitoring rice grain sieve losses in combine harvesters. *Biosyst. Eng.* **2016**, *147*, 51–66. [CrossRef]
119. Guan, C.; Fu, J.; Xu, L.; Jiang, X.; Wang, S.; Cui, Z. Study on the reduction of soil adhesion and tillage force of bionic cutter teeth in secondary soil crushing. *Biosyst. Eng.* **2022**, *213*, 133–147. [CrossRef]
120. Han, J.; Yan, X.; Tang, H. Method of controlling tillage depth for agricultural tractors considering engine load characteristics. *Biosyst. Eng.* **2023**, *227*, 95–106. [CrossRef]
121. Ma, Z.; Wu, Z.; Li, Y.; Song, Z.; Yu, J.; Li, Y.; Xu, L. Study of the grain particle-conveying performance of a bionic non-smooth-structure screw conveyor. *Biosyst. Eng.* **2024**, *238*, 94–104. [CrossRef]
122. Li, Z.; Zhou, S.; Xiao, Y.; Wang, L. Sensorless vector control of permanent magnet synchronous linear motor based on self-adaptive super-twisting sliding mode controller. *IEEE Access* **2019**, *7*, 44998–45011. [CrossRef]
123. Zhang, Y.; Jin, J.; Huang, L. Model-Free Predictive Current Control of PMSM Drives Based on Extended State Observer Using Ultralocal Model. *IEEE Trans. Ind. Electron.* **2021**, *68*, 993–1003. [CrossRef]
124. Zhang, Y.; Shen, W.; Yang, H. An Improved Deadbeat Predictive Current Control of PMSM Drives Based on the Ultra-local Model. *Chin. J. Integr. Med.* **2023**, *9*, 27–37. [CrossRef]
125. Su, G.; Li, H.; Li, Z.; Zhou, Y. Model-Free Control of Permanent Magnet Synchronous Linear Motor Based on Ultra-local Model. *Chin J Integr Med.* **2024**, *10*, 73–82. [CrossRef]
126. Safaei, A.; Mahyuddin, M.N. Adaptive Model-Free Control Based on an Ultra-Local Model With Model-Free Parameter Estimations for a Generic SISO System. *IEEE Access* **2018**, *6*, 4266–4275. [CrossRef]
127. Chen, J.; Fan, Y.; Cheng, M.; Zhang, Q.; Chen, Q. Parameter-free ultralocal model-based deadbeat predictive current control for PMVMs using finite-time gradient method. *IEEE Trans. Ind. Electron.* **2023**, *70*, 5549–5559. [CrossRef]
128. Yuan, X.; Zuo, Y.; Fan, Y.; Lee, C.H.T. Model-free predictive current control of SPMSM drives using extended state observer. *IEEE Trans. Ind. Electron.* **2022**, *69*, 6540–6550. [CrossRef]
129. Long, J.; Yang, M.; Chen, Y.; Liu, K.; Xu, D. Current-Controller-Free Self-Commissioning Scheme for Deadbeat Predictive Control in Parametric Uncertain SPMSM. *IEEE Access* **2021**, *9*, 289–302. [CrossRef]

130. Brosch, A.; Hanke, S.; Wallscheid, O.; Bocker, J. Data-Driven Recursive Least Squares Estimation for Model Predictive Current Control of Permanent Magnet Synchronous Motors. *IEEE Trans. Power Electron.* **2021**, *36*, 2179–2190. [CrossRef]
131. Siami, M.; Khaburi, D.A.; Abbaszadeh, A.; Rodriguez, J. Robustness Improvement of Predictive Current Control Using Prediction Error Correction for Permanent-Magnet Synchronous Machines. *IEEE Trans. Ind. Electron.* **2016**, *63*, 3458–3466. [CrossRef]
132. Akter, M.P.; Mekhilef, S.; Tan, N.M.L.; Akagi, H. Modified model predictive control of a bidirectional AC-DC converter based on Lyapunov function for energy storage systems. *IEEE Trans. Ind. Electron.* **2016**, *67*, 704–715. [CrossRef]
133. Brosch, A.; Wallscheid, O.; Bocker, J. Long-Term Memory Recursive Least Squares Online Identification of Highly Utilized Permanent Magnet Synchronous Motors for Finite-Control-Set Model Predictive Control. *IEEE Trans. Power Electron.* **2023**, *38*, 1451–1467. [CrossRef]
134. Sun, J.; Gong, J.; Chen, B.; Zha, X. Analysis and design of repetitive controller based on regeneration spectrum and sensitivity function in active power filter system. *IET Power Electron.* **2014**, *67*, 2133–2140. [CrossRef]
135. Cao, G.Z.; Zhou, H.H.; Huang, S.D.; Hu, H.J.; He, J.B. Position Estimation Method of a Permanent Magnet Synchronous Motor Based on Moving Horizon Estimation EKF Algorithm. *IEEE Trans. Magn.* **2025**, *61*, 8201106. [CrossRef]
136. Wang, Y.; Liao, W.; Huang, S.; Zhang, J.; Yang, M.; Li, C.; Huang, S. A Robust DPCC for IPMSM Based on a Full Parameter Identification Method. *IEEE Trans. Ind. Electron.* **2023**, *70*, 7695–7705. [CrossRef]
137. Wang, Q.; Zhao, X.; Yang, P.; Hua, W.; Buja, G. Effects of Triangular Wave Injection and Current Differential Terms on Multiparameter Identification for PMSM. *IEEE Trans. Power Electron.* **2024**, *39*, 2943–2947. [CrossRef]
138. Wang, Q.; Wang, G.; Zhao, N.; Zhang, G.; Cui, Q.; Xu, D.G. An Impedance Model-Based Multiparameter Identification Method of PMSM for Both Offline and Online Conditions. *IEEE Trans. Power Electron.* **2021**, *36*, 727–738. [CrossRef]
139. Zhang, Q.; Guo, H.; Liu, Y.; Guo, C.; Zhang, F.; Zhang, Z.; Li, G. A Novel Error-Injected Solution for Compensation of Current Measurement Errors in PMSM Drive. *IEEE Trans. Ind. Electron.* **2023**, *70*, 4608–4619. [CrossRef]
140. Wang, F.; Ke, D.; Yu, X.; Huang, D. Enhanced predictive model based deadbeat control for PMSM drives using exponential extended state observer. *IEEE Trans. Ind. Electron.* **2022**, *69*, 2357–2369. [CrossRef]
141. Wang, F.; Kong, W.; Qu, R. Model Parameter Self-Correcting Deadbeat Predictive Current Control for SPMSM Drives. *IEEE Trans. Ind. Electron.* **2025**, *72*, 2357–2368. [CrossRef]
142. Sun, X.; Zhu, Y.; Cai, Y.; Yao, M.; Sun, Y.; Lei, G. Optimized-Sector-Based Model Predictive Torque Control With Sliding Mode Controller for Switched Reluctance Motor. *IEEE Trans. Energy Convers.* **2024**, *39*, 379–388. [CrossRef]
143. Zhang, X.; Fang, S.; Zhang, H. Predictive Current Error Compensation Based Strong Robust Model Predictive Control for PMSM Drive Systems. *IEEE Trans. Ind. Electron.* **2024**, *71*, 15280–15289. [CrossRef]
144. Zhang, X.; Cao, Y.; Zhang, C.; Niu, S. Model Predictive Control for PMSM Based on the Elimination of Current Prediction Errors. *IEEE J. Emerg. Sel. Top. Power Electron.* **2024**, *12*, 2651–2660. [CrossRef]
145. Chen, P.; Ma, R.; Bai, H.; Chen, Z.; Song, S. Rotating High Frequency Voltage Injection Strategy Considering Time Delay for PMSM Based on Dual Heterodyne Method. *IEEE Trans. Energy Convers.* **2025**, *40*, 901–910. [CrossRef]
146. Han, Y.; Gong, C.; Yan, L.; Wen, H.; Wang, Y.; Shen, K. Multiobjective Finite Control Set Model Predictive Control Using Novel Delay Compensation Technique for PMSM. *IEEE Trans. Power Electron.* **2020**, *35*, 11193–11204. [CrossRef]
147. Wu, T.; Luo, D.; Wu, X.; Liu, K.; Huang, S.; Peng, X. Square-Wave Voltage Injection Based PMSM Sensorless Control Considering Time Delay at Low Switching Frequency. *IEEE Trans. Ind. Electron.* **2022**, *69*, 5525–5535. [CrossRef]
148. Zhang, X.; Cheng, Y.; Zhao, Z.; Yan, K. Optimized model predictive control with dead-time voltage vector for PMSM drives. *IEEE Trans. Power Electron.* **2021**, *36*, 3149–3158. [CrossRef]
149. Sun, X.; Cao, J.; Lei, G.; Guo, Y.; Zhu, J. A robust deadbeat predictive controller with delay compensation based on composite sliding mode observer for PMSMs. *IEEE Trans. Power Electron.* **2021**, *36*, 10742–10752. [CrossRef]
150. Bouafia, A.; Gaubert, J.-P.; Krim, F. Predictive Direct Power Control of Three-Phase Pulsewidth Modulation (PWM) Rectifier Using Space-Vector Modulation (SVM). *IEEE Trans. Power Electron.* **2010**, *25*, 228–236. [CrossRef]
151. Moon, H.-T.; Kim, H.-S.; Youn, M.-J. A discrete-time predictive current control for PMSM. *IEEE Trans. Power Electron.* **2003**, *18*, 464–472. [CrossRef]
152. Sun, X.; Xu, Z.; Pan, M.; Sun, C.; Pan, W.; Lei, G. Sensorless Control Strategy for SRM Based on Flux Linkage in Medium to High-Speed Range. *IEEE Trans. Ind. Electron.* 2025; *early access*. [CrossRef]
153. Zhang, X.; Wang, Z.; Zhao, Z.; Cheng, M. Model Predictive Voltage Control for SPMSM Drives with Parameter Robustness Optimization. *IEEE Trans. Transp. Electrific.* **2022**, *8*, 3151–3163. [CrossRef]
154. Yang, M.; Lang, X.; Long, J.; Xu, D. Flux Immunity Robust Predictive Current Control with Incremental Model and Extended State Observer for PMSM Drive. *IEEE Trans. Power Electron.* **2017**, *32*, 9267–9279. [CrossRef]
155. He, L.; Wang, F.; Wang, J.; Rodriguez, J. Zynq implemented Luenberger disturbance observer based predictive control scheme for PMSM drives. *IEEE Trans. Power Electron.* **2020**, *35*, 1770–1778. [CrossRef]
156. Wang, H.; Gan, C.; Zhang, C.; Ren, H.; Qu, R. Parameter Robust Predictive Current Control for PMSM Drives Based on Self-Tuning Incremental Model and Voltage Constraint Compensation. *IEEE Trans. Power Electron.* **2025**, *40*, 11268–11282. [CrossRef]

157. Sun, X.; Lin, X.; Zhang, L.; Yao, M.; Cai, Y. An improved torque enhancement strategy for dual three-phase PMSM based on model-free predictive current control. *IEEE Trans. Transport. Electrific.* **2025**, *11*, 176–187. [CrossRef]
158. Fuentes, E.J.; Silva, C.A.; Yuz, J.I. Predictive Speed Control of a Two-Mass System Driven by a Permanent Magnet Synchronous Motor. *IEEE Trans. Ind. Electron.* **2012**, *59*, 2840–2848. [CrossRef]
159. Gao, X.; Abdelrahem, M.; Hackl, C.M.; Zhang, Z.; Kennel, R. Direct Predictive Speed Control with a Sliding Manifold Term for PMSM Drives. *IEEE J. Emerg. Sel. Top. Power Electron.* **2020**, *8*, 1258–1267. [CrossRef]
160. Sun, X.; Su, Z.; Lei, G.; Yao, M. Robust predictive cascaded speed and current control for PMSM drives considering parameter variations. *IEEE Trans. Ind. Electron.* **2024**, *71*, 10235–10245. [CrossRef]
161. Wang, T.; Luo, G.; Chen, Z.; Tu, W.; Liu, C. An improved robust model predictive speed control with inertia identification for PMSM drives in electro-hydrostatic actuator. *IEEE Trans. Power Electron.* **2023**, *38*, 13825–13841. [CrossRef]
162. Li, T.; Sun, X.; Yang, Z.; Lei, G. Simplified two-step model predictive control with fast voltage vector search. *IEEE Trans. Ind. Electron.* **2025**, *72*, 3303–3312. [CrossRef]
163. Li, R.; Huang, D.; Chen, Y.; Zhao, Q. Deadbeat Predictive Speed Control of PMSM via Enhanced Low-Gain Disturbance Observer Design and Inertia Identification. *IEEE J. Emerg. Sel. Top. Power Electron.* **2025**, *13*, 2524–2534. [CrossRef]
164. Sun, J.; He, X.; Ge, X.; Wu, X.; Shen, J.; Song, Y. Detection of key organs in tomato based on deep migration learning in a complex background. *Agriculture* **2018**, *8*, 196. [CrossRef]
165. Peng, Y.; Zhao, S.; Liu, J. Fused deep features-based grape varieties identification using support vector machine. *Agriculture* **2021**, *11*, 869. [CrossRef]
166. Ji, K.; Li, Y.; Liang, Z.; Liu, Y.; Cheng, J.; Wang, H.; Zhu, R.; Xia, S.; Zheng, G. Device and method suitable for matching and adjusting reel speed and forward speed of multi-crop harvesting. *Agriculture* **2022**, *12*, 213. [CrossRef]
167. Zhou, H.; Jia, W.; Li, Y.; Ou, M. Method for estimating canopy thickness using ultrasonic sensor technology. *Agriculture* **2021**, *11*, 1011. [CrossRef]
168. Zhang, B.; Chen, X.; Liang, R.; Wang, X.; Meng, H.; Kan, Z. Calibration and test of contact parameters between chopped cotton stalks using response surface methodology. *Agriculture* **2022**, *12*, 1851. [CrossRef]
169. Qiu, S.; Yu, Y.; Feng, Y.; Tang, Z.; Cui, Q.; Yuan, X. Crushing characteristics of sorghum grains subjected to compression and impact loading at different moisture contents. *Agriculture* **2022**, *12*, 1422. [CrossRef]
170. Zhu, Q.; Zhang, H.; Zhu, Z.; Gao, Y.; Chen, L. Structural design and simulation of pneumatic conveying line for a paddy side-deep fertilisation system. *Agriculture* **2022**, *12*, 867. [CrossRef]
171. Gao, J.; Zhang, J.; Lu, D. Design and atomization experiments of an ultrasonic atomizer with a levitation mechanism. *Appl. Eng. Agric.* **2016**, *32*, 353–360. [CrossRef]
172. Kumi, F.; Mao, H.; Li, Q.; Luhua, H. Assessment of tomato seedling substrate-root quality using X-ray computed tomography and scanning electron microscopy. *Appl. Eng. Agric.* **2016**, *32*, 417–427. [CrossRef]
173. Shi, Q.; Mao, H.; Guan, X. Numerical simulation and experimental verification of the deposition concentration of an unmanned aerial vehicle. *Appl. Eng. Agric.* **2019**, *35*, 367–376. [CrossRef]
174. Fordjour, A.; Zhu, X.; Yuan, S.; Dwomoh, F.A.; Issaka, Z. Numerical simulation and experimental study on internal flow characteristic in the dynamic fluidic sprinkler. *Appl. Eng. Agric.* **2020**, *36*, 61–70. [CrossRef]
175. Hua, L.; Li, H.; Jiang, Y. Axis-switching behavior of liquid jets issued from non-circular nozzles under low-intermediate pressure. *Appl. Eng. Agric.* **2021**, *37*, 367–378. [CrossRef]
176. Sun, X.; Chen, Z.; Pan, M.; Cai, Y.; Jin, Z.; Lei, G.; Tian, X. Robust energy management optimization for PHEB considering driving uncertainties by using sequential Taguchi method. *IEEE Trans. Transport. Electrific.* **2025**, *11*, 5191–5200. [CrossRef]

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